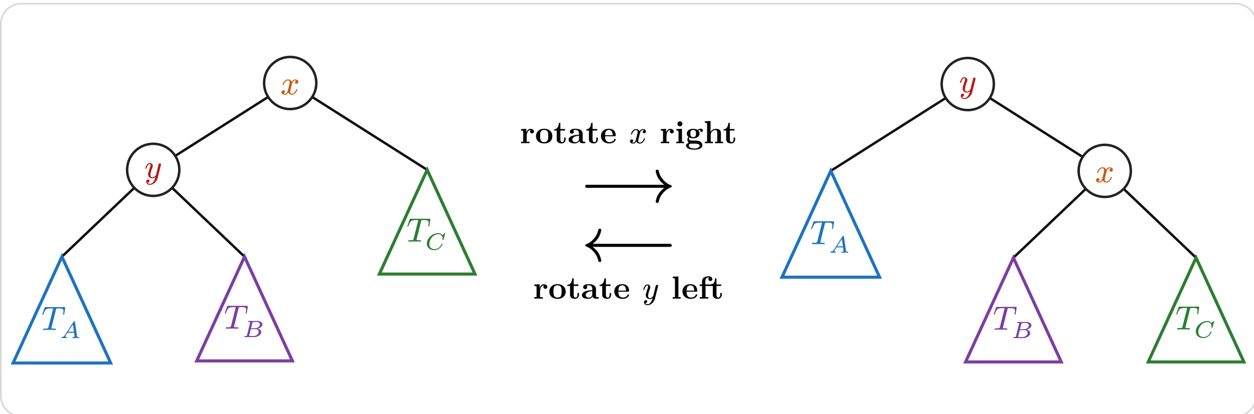
typed-dsa

Declarative data-structure diagrams for Typst

User Guide



Version 0.2.0

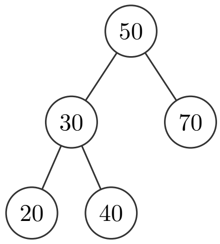
Contents

1 Introduction

typed-dsa renders data-structure diagrams from a declarative description inside Typst documents. It is built on top of CeTZ. The package targets CS coursework: lecture notes, problem sets, exam solutions, where you need to show how an **operation** changes a structure, not just draw it once.

This guide covers every builder, argument, and operation, each with a runnable example: source on the left, rendered output on the right.

```typst
#bst(50, 30, 70, 20, 40).diagram
```



2 Getting started

2.1 Installation and import

Import the package from the Typst preview namespace. A wildcard import gives you every public symbol:

```typst
#import "@preview/typed-dsa:0.1.0": *
```

Or import only what you need:

```typst
#import "@preview/typed-dsa:0.1.0": bst, avl, min-heap, max-heap
```

Note. A wildcard import shadows Typst's built-in `stack` function. Use `std.stack` for layout, or import `typed-dsa` selectively to avoid the name clash.

The package exports these symbols:

Symbol	Purpose
`bst` , `avl`	Binary search tree / AVL tree builders.
`min-heap` , `max-heap`	Binary heap builders.
`linked-list` , `doubly-linked-list` , `stack` , `queue`	Linear-structure builders.
`array-view` , `matrix`	Static array and matrix/grid builders.
`merge-sort` , `merge-operation` , `partition-step` , `quick-sort` , `bubble-sort` , `insertion-sort` , `selection-sort`	Automatic

	sorting traces.
`graph`	Graph builder, from an adjacency dict.
`tree` , `node` , `subtree`	Hand-composed abstract trees.
`transition`	One-shot before → after operation diagrams for trees and heaps.
`tree-insert` , `tree-delete` , `tree-search`	Tree operation constructors, for `transition` .

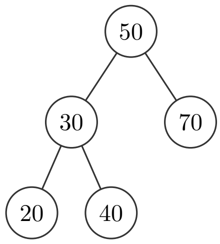
`heap-insert` , `heap-extract`	Heap operation constructors, for `transition` .
`sequence`	Wrapped layout for operation-step sequences.
`op-arrow`	Reusable labeled arrow, for custom operation layouts.
`theme` , `resolve`	The default style dictionary and its merge helper.

2.2 Your first diagram

Every builder takes keys and returns an object; `.diagram` is its drawing.

```typst
#bst(50, 30, 70, 20, 40).diagram
```

Typst



3 Structures

Every structure builder returns a **unified object**: a dictionary that is both the drawing (`.diagram`) and the thing you operate on. Calling an operation field returns a **step**; Section 6 covers that model in full.

This section documents each builder's own arguments and methods. Section 6 covers the step shape they share.

3.1 `bst()` and `avl()`

Full signatures:

```typst
#bst(..keys, style: (:), edge-customizations: (), node-customizations: (), node-labels: (:))
#avl(..keys, style: (:), edge-customizations: (), node-customizations: (), node-labels: (:))
```

Both build a binary tree by inserting `keys` one at a time, in the order given. `bst` is a plain binary search tree; `avl` additionally rebalances after every insertion and deletion, so the tree stays height-balanced.

Argument	Type	Default	Description
`..keys`	`int` / `content`	(required)	Keys inserted into the tree in order, one BST/AVL insert per key. Duplicate keys are ignored.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`edge-customizations`	`array`	`()`	`(parent, child, options)` tuples restyling individual tree edges by node value. See Section 3.11.
`node-customizations`	`array`	`()`	`(node, options)` tuples restyling individual tree nodes by value.
`node-labels`	`array` / `dictionary`	`(:)`	Labels drawn outside individual tree nodes.

Nested keys for `style`

Path	Type	Default	Description
`style.node-radius`	`float`	`0.34`	Circle radius, or half the square/diamond/hexagon size, in canvas units.
`style.node-shape`	`str`	`"circle"`	`"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.x-gap`	`float`	`1.05`	Horizontal spacing between in-order columns.
`style.y-gap`	`float`	`1.2`	Vertical spacing between depths.
`style.node-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Node outline.
`style.node-fill`	`color`	`white`	Default node fill.

`style.edge-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Parent-child edge stroke.
`style.edge-arrow`	`none` / `bool` / `str`	`none`	Edge arrowheads: `none` / `false` , `"end"` or `true` , `"start"` , or `"both"` .
`style.edge-pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`style.edge-wave-amplitude`	`float`	`0.07`	Wave height in canvas units.
`style.edge-wave-step`	`float`	`0.14`	Approximate wavelength in canvas units.
`style.node-text`	`dictionary`	`(size: 9pt)`	Node text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Annotation text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .

`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Nested keys for `edge-customizations`

Path	Type	Default	Description
`edge-customizations[]`	`tuple`	n/a	One `(from, to, options)` tuple. For trees, `from` / `to` are parent and child values; for graphs, node labels.
`edge-customizations[].from`	`content`	required	Source node identity.
`edge-customizations[].to`	`content`	required	Target node identity.
`edge-customizations[].options`	`dictionary`	required	Per-edge options.
`edge-customizations[].options.stroke`	`stroke`	none	Full stroke override. Wins over `color` if both are set.
`edge-customizations[].options.color`	`color`	none	Paint only, at the default thickness.
`edge-customizations[].options.pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`edge-customizations[].options.arrow`	`none` / `bool` / `str`	none	Per-edge arrowheads: `none` , `"end"` / `true` , `"start"` , or `"both"` .
`edge-customizations[].options.bend`	`str` / `bool`	`false`	`"left"` or `"right"` curves the edge relative to its travel direction. `false` keeps it straight.
`edge-customizations[].options.angle`	`angle`	`25deg`	How sharply a bent edge curves. Ignored when `bend` is `false` .
`edge-customizations[].options.label`	`content` / `dictionary`	none	Tree edge label content, or graph edge-label text overrides. A dictionary can include `content` for trees.

Nested keys for `edge-customizations[].options.label`

Path	Type	Default	Description
`edge-customizations[].options.label.size`	`length`	`style.label-text.size`	Edge-label size.
`edge-customizations[].options.label.content`	`content`	tree only	Tree edge label content. Graph edge labels come from adjacency entries such as `("B", [4])` .
`edge-customizations[].options.label.color`	`color`	`style.label-text.color`	Edge-label text color.
`edge-customizations[].options.label.rotation`	`angle` / `str`	`style.label-text.rotation`	An angle, or `"edge"` to follow the tree/graph edge angle.
`edge-customizations[].options.label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`edge-customizations[].options.label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `node-customizations`

Path	Type	Default	Description
`node-customizations[]`	`tuple`	n/a	One `(node, options)` tuple.
`node-customizations[].node`	`content`	required	For `bst` / `avl` , the inserted key.
`node-customizations[].options`	`dictionary`	required	Per-node drawing options.
`node-customizations[].options.fill`	`color`	normal fill	Node fill.
`node-customizations[].options.stroke`	`stroke`	normal stroke	Node outline.
`node-customizations[].options.shape`	`str`	`style.node-shape`	Trees/graphs only: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`node-customizations[].options.node-radius`	`float`	`style.node-radius`	Trees/graphs only. Shape size.
`node-customizations[].options.text`	`dictionary`	`style.node-text`	Text style merged into the node text.

Nested keys for `node-labels`

Path	Type	Default	Description
`node-labels[]`	`tuple`	n/a	One `(node, label)` tuple.
`node-labels[].node`	`content`	required	For `bst` / `avl` , the inserted key.
`node-labels[].label`	`content` / `dictionary`	required	Bare content for quick labels, or a dictionary with `content` plus style and placement keys.
`node-labels[].label.content`	`content`	required for dict labels	Label body. `body` is also accepted.
`node-labels[].label.position`	`str` / `angle`	`"right"`	`"right"` , `"left"` , `"top"` , `"bottom"` , or an angle.
`node-labels[].label.offset`	`point`	`(0, 0)`	Extra `(dx, dy)` shift after the position is applied.
`node-labels[].label.size`	`length`	`style.label-text.size`	Node-label size.

`node-labels[].label.color`	`color`	`style.label-text.color`	Node-label text color.
`node-labels[].label.rotation`	`angle`	`style.label-text.rotation`	Rotation of the label text itself.
`node-labels[].label.font`	`str / array`	`style.label-text.font`	Typst text font selection.
`node-labels[].label.weight`	`str / int`	`style.label-text.weight`	Typst text weight.

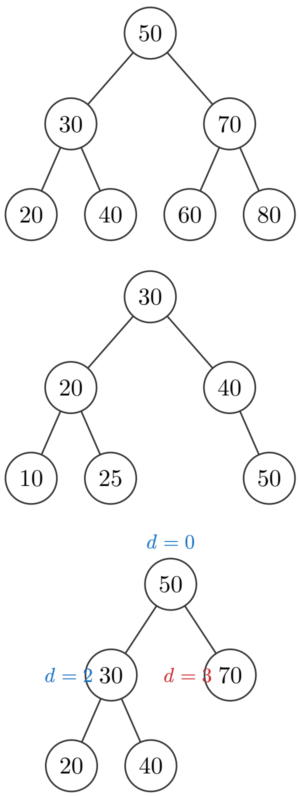
Nested keys for `style.node-labels`

Path	Type	Default	Description
`style.node-labels.position`	`str / angle`	`"right"`	Default position for all node labels in this diagram.
`style.node-labels.offset`	`point`	`(0, 0)`	Default extra `(dx, dy)` shift for all node labels.
`style.node-labels.gap`	`float`	`0.22`	Default distance from the node boundary in canvas units.
`style.node-labels.size`	`length`	`style.label-text.size`	Default node-label text size.
`style.node-labels.color`	`color`	`style.label-text.color`	Default node-label text color.
`style.node-labels.font`	`str / array`	inherits	Typst text font selection.
`style.node-labels.weight`	`str / int`	inherits	Typst text weight.
`style.node-labels.rotation`	`angle`	`style.label-text.rotation`	Default rotation of the label text itself.

Field	Call	Effect
`.diagram`	n/a	The rendered tree.
`.insert`	`(obj.insert)(key)`	Insert `key` . Returns a step (Section 6).
`.delete`	`(obj.delete)(key)`	Delete `key` . AVL deletes rebalance when removal makes a subtree too heavy. Returns a step.
`.search`	`(obj.search)(key)`	Highlight the traversal path to `key` . Returns a step.

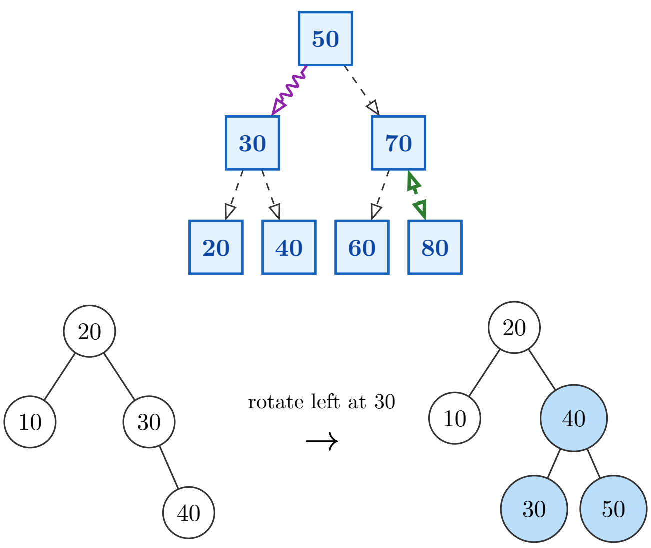
```typst
#bst(50, 30, 70, 20, 40, 60, 80).diagram
#avl(10, 20, 30, 40, 50, 25).diagram

#bst(
  50, 30, 70, 20, 40,
  node-labels: (
    (50, (content: [$d=0$], position: "top")),
    (30, [$d=2$]),
    (70, (content: [$d=3$], color: rgb("#C92A2A"))),
  ),
  style: (node-labels: (position: "left", color: rgb("#1971C2"))),
).diagram
```



Argument coverage: this example combines tree-level styling, nested `node-text` , edge arrows/patterns, per-edge customizations, and diff-highlight dictionaries.

```typst
#std.stack(spacing: 1em,
  bst(
    50, 30, 70, 20, 40, 60, 80,
    style: (
      node-shape: "square",
      node-radius: 0.38,
      node-fill: rgb("#E3F2FD"),
      node-stroke: 1pt + rgb("#1565C0"),
      node-text: (size: 10pt, weight: "bold", color: rgb("#0D47A1")),
      edge-arrow: "end",
      edge-pattern: "dashed",
      scale: 0.92,
      y-gap : 1.5,
    ),
    edge-customizations: (
      (50, 30, (pattern: "wavy", color: rgb("#8E24AA"))),
      (70, 80, (stroke: 1.4pt + rgb("#2E7D32"), arrow: "both")),
    ),
  ).diagram,
  transition(
    "avl",
    (20, 10, 30, 40),
    tree-insert(50),
    style: (
      new-style: (shape: "square", stroke: 1.5pt + rgb("#2E7D32")),
      path-style: (stroke: 1.2pt + rgb("#F9A825")),
      rotate-style: (node-radius: 0.44),
    ),
  ),
)
```



3.2 `min-heap()` and `max-heap()`

Full signatures:

```typst
#min-heap(..keys, style: (:))
#max-heap(..keys, style: (:))
```

A heap is an array underneath. Index `i`'s children live at `2i+1` and `2i+2`. Each key sifts up as it's inserted, and the diagram draws the complete binary tree that array shape forms. `min-heap` keeps the smallest key at the root; `max-heap` keeps the largest.

Argument	Type	Default	Description
`..keys`	`int` / `content`	(required)	Keys inserted into the heap in order, one sift-up per key.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.

Nested keys for `style`

Path	Type	Default	Description
`style.node-radius`	`float`	`0.34`	Circle radius, or half the square/diamond/hexagon size, in canvas units.
`style.node-shape`	`str`	`"circle"`	`"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.x-gap`	`float`	`1.05`	Horizontal spacing between in-order columns.
`style.y-gap`	`float`	`1.2`	Vertical spacing between depths.
`style.node-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Node outline.
`style.node-fill`	`color`	`white`	Default node fill.
`style.edge-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Parent-child edge stroke.
`style.edge-arrow`	`none` / `bool` / `str`	`none`	Edge arrowheads: `none` / `false` , `"end"` or `true` , `"start"` , or `"both"` .
`style.edge-pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`style.edge-wave-amplitude`	`float`	`0.07`	Wave height in canvas units.
`style.edge-wave-step`	`float`	`0.14`	Approximate wavelength in canvas units.
`style.node-text`	`dictionary`	`(size: 9pt)`	Node text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Annotation text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

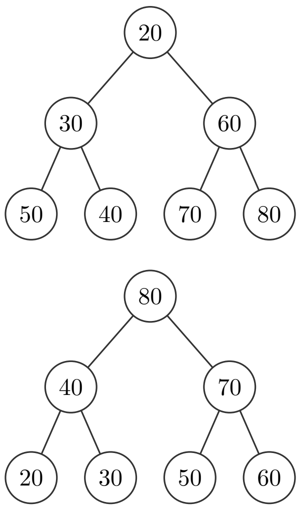
Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)`.
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"`, `"square"`, `"diamond"`, or `"hexagon"`.
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size`, `color`, `font`, and `weight`.

Field	Call	Effect

`.diagram`	n/a	The rendered heap.
`.insert`	`(obj.insert)(key)`	Append `key` and sift it up. Returns a step.
`.extract`	`(obj.extract)()`	Remove the root (no key), move the last leaf up, and sift it down. Returns a step.

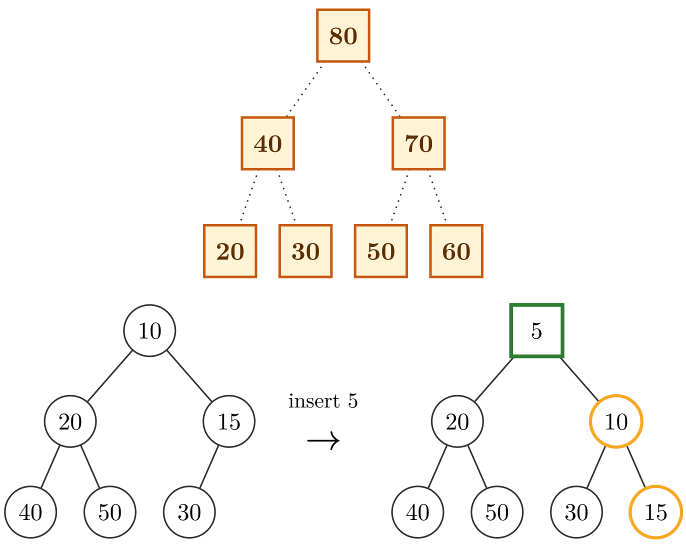
Typst

```
#min-heap(50, 30, 70, 20, 40, 60, 80).diagram
#max-heap(50, 30, 70, 20, 40, 60, 80).diagram
```



Argument coverage: heaps reuse tree styling, and heap operations use the same diff-highlight dictionaries.

```typst
#std.stack(spacing: 1em,
  max-heap(
    50, 30, 70, 20, 40, 60, 80,
    style: (
      node-shape: "square",
      node-radius: 0.36,
      y-gap: 1.5,
      node-fill: rgb("#FFF3D6"),
      node-stroke: 1pt + rgb("#C75B12"),
      node-text: (size: 10pt, color: rgb("#5D2F00"), weight: "bold"),
      edge-pattern: "dotted",
      scale: 0.95,
    ),
  ).diagram,
  transition(
    "min-heap",
    (10, 20, 15, 40, 50, 30),
    heap-insert(5),
    style: (
      diff-colors: false,
      new-style: (shape: "square", stroke: 1.4pt + rgb("#2E7D32")),
      path-style: (stroke: 1.2pt + rgb("#F9A825")),
    ),
  ),
)
```



Note. A binary heap supports inserting and pulling out the root efficiently. It has no key-based `delete` or `search` the way `bst` / `avl` do: that's not a gap in typed-dsa, it's what a heap is.

3.3 `linked-list()`

Full signature:

```typst
#linked-list(
  ..vals,
  style: (:),
  pointer: false,
  addresses: none,
  head: false,
)
```

Argument	Type	Default	Description
`..vals`	`int` / `content`	(required)	Values in the list, head first.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`pointer`	`bool`	`false`	Draw each node as a `data | next` cell pair instead of a single value cell.
`addresses`	`none` / `array`	`none`	With `pointer: true` , one address label drawn under each node, in order.
`head`	`bool`	`false`	Draw a “Head” arrow pointing at the first node.

Nested keys for `style`

Path	Type	Default	Description
`style.box-w`	`float`	`0.95`	Cell width.
`style.box-h`	`float`	`0.7`	Cell height.
`style.box-gap`	`float`	`0.55`	Gap between cells or list nodes.
`style.box-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Cell outline.
`style.box-fill`	`color`	`white`	Default cell fill.
`style.ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `linked-list(pointer: true)` .
`style.prev-ptr-fill`	`color`	`rgb("#D7ECC9")`	Previous-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.next-ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.node-text`	`dictionary`	`(size: 9pt)`	Cell text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Head, address, front, and rear label text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .

`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

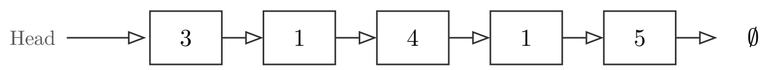
Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)`.
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"`, `"square"`, `"diamond"`, or `"hexagon"`.
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.

`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Field	Call	Effect
`.diagram`	n/a	The rendered list.
`.insert`	`(obj.insert)(v)`	Append `v` at the tail. Returns a step.
`.delete`	`(obj.delete)(v)`	Remove the first node equal to `v` . Returns a step.

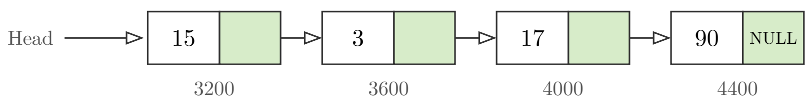
```typst
#linked-list(3, 1, 4, 1, 5, head: true).diagram
```

Typst



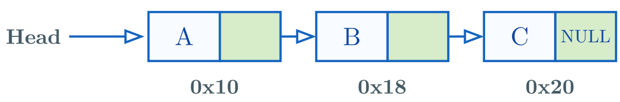

```typst
#linked-list(15, 3, 17, 90, pointer: true, head: true,
  addresses: ("3200", "3600", "4000", "4400")).diagram
```

Typst



Argument coverage: `pointer` , `addresses` , `head` , label text, pointer-cell fills, scaling, and operation mark styling.

```typst
#let xs = linked-list(
  [A], [B], [C],
  pointer: true,
  head: true,
  addresses: ("0x10", "0x18", "0x20"),
  style: (
    box-w: 0.9,
    box-h: 0.62,
    box-gap: 0.45,
    box-fill: rgb("#F8FBFF"),
    box-stroke: 0.8pt + rgb("#1565C0"),
    ptr-fill: rgb("#D7ECC9"),
    node-text: (size: 9pt, color: rgb("#0D47A1")),
    label-text: (size: 7.5pt, color: rgb("#455A64"), weight: "bold"),
    scale: 1.05,
    new-style: (fill: rgb("#C8E6C9"), stroke: 1pt + rgb("#2E7D32")),
    remove-style: (fill: rgb("#FFCDD2"), stroke: 1pt + rgb("#C62828")),
  ),
)
#xs.diagram
```



3.4 `doubly-linked-list()`

Full signature:

```typst
#doubly-linked-list(
  ..vals,
  style: (:),
  pointer: false,
  addresses: none,
  head: false,
)
```

Argument	Type	Default	Description
`..vals`	`int` / `content`	(required)	Values in the list, head first.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`pointer`	`bool`	`false`	When `false` , draw one value cell per node with a forward arrow above and

			a backward arrow below. When `true` , draw `prev | data | next` cells.
`addresses`	`none` / `array`	`none`	With `pointer: true` , one address label drawn under each node, in order.
`head`	`bool`	`false`	Draw a "Head" arrow pointing at the first node.

Nested keys for `style`

Path	Type	Default	Description
`style.box-w`	`float`	`0.95`	Cell width.
`style.box-h`	`float`	`0.7`	Cell height.
`style.box-gap`	`float`	`0.55`	Gap between cells or list nodes.
`style.box-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Cell outline.
`style.box-fill`	`color`	`white`	Default cell fill.
`style.ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `linked-list(pointer: true)` .
`style.prev-ptr-fill`	`color`	`rgb("#D7ECC9")`	Previous-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.next-ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.node-text`	`dictionary`	`(size: 9pt)`	Cell text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Head, address, front, and rear label text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description

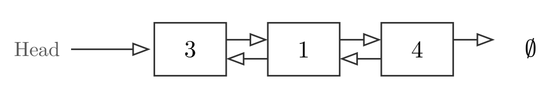
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

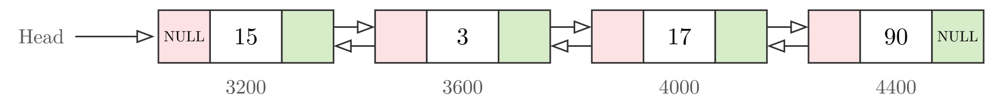
Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Field	Call	Effect
`.diagram`	n/a	The rendered list.
`.insert`	`(obj.insert)(v)`	Append `v` at the tail. Returns a step.
`.delete`	`(obj.delete)(v)`	Remove the first node equal to `v` . Returns a step.

```typst
#doubly-linked-list(3, 1, 4, head: true).diagram
```

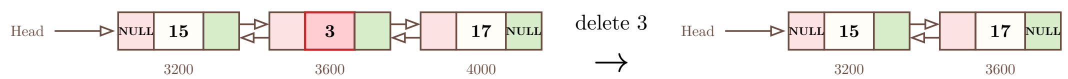


```typst
#doubly-linked-list(15, 3, 17, 90, pointer: true, head: true,
  addresses: ("3200", "3600", "4000", "4400"),
  style: (
    prev-ptr-fill: rgb("#FDE2E4"),
    next-ptr-fill: rgb("#D7ECC9"),
  )).diagram
```



Argument coverage: doubly linked pointer cells have separate previous and next pointer fills, plus the same address, head, label, and mark styling as `linked-list` .

```typst
#let xs = doubly-linked-list(
  15, 3, 17,
  pointer: true,
  head: true,
  addresses: ("3200", "3600", "4000"),
  style: (
    box-w: 0.82,
    box-h: 0.62,
    box-gap: 0.5,
    box-fill: rgb("#FFFDF7"),
    box-stroke: 0.8pt + rgb("#6D4C41"),
    prev-ptr-fill: rgb("#FDE2E4"),
    next-ptr-fill: rgb("#D7ECC9"),
    node-text: (size: 8.5pt, weight: "bold"),
    label-text: (size: 7pt, color: rgb("#6D4C41")),
    path-style: (fill: rgb("#FFE9A8"), stroke: 1pt + rgb("#F9A825")),
    remove-style: (fill: rgb("#FFCDD2"), stroke: 1pt + rgb("#C62828")),
    scale : 0.8,
  ),
)
#(xs.delete)(3).diagram
```



3.5 `stack()`

Full signature:

```typst
#stack(..vals, style: (:), top-label: [top])
```

Argument	Type	Default	Description

`..vals`	`int` / `content`	(required)	Values in the stack; the **first** argument is the top.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`top-label`	`content`	`[top]`	Label beside the top cell.

Nested keys for `style`

Path	Type	Default	Description
`style.box-w`	`float`	`0.95`	Cell width.
`style.box-h`	`float`	`0.7`	Cell height.
`style.box-gap`	`float`	`0`	Gap between cells or list nodes.
`style.box-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Cell outline.
`style.box-fill`	`color`	`white`	Default cell fill.
`style.ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `linked-list(pointer: true)` .
`style.prev-ptr-fill`	`color`	`rgb("#D7ECC9")`	Previous-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.next-ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.node-text`	`dictionary`	`(size: 9pt)`	Cell text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Head, address, front, and rear label text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.

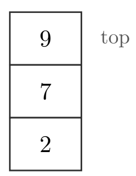
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

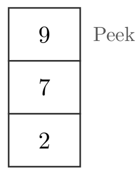
Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Field	Call	Effect
`.diagram`	n/a	The rendered stack.
`.push`	`(obj.push)(v)`	Push `v` onto the top. Returns a step.
`.pop`	`(obj.pop)()`	Pop the top value (no argument). Returns a step.

```typst
#stack(9, 7, 2).diagram
```

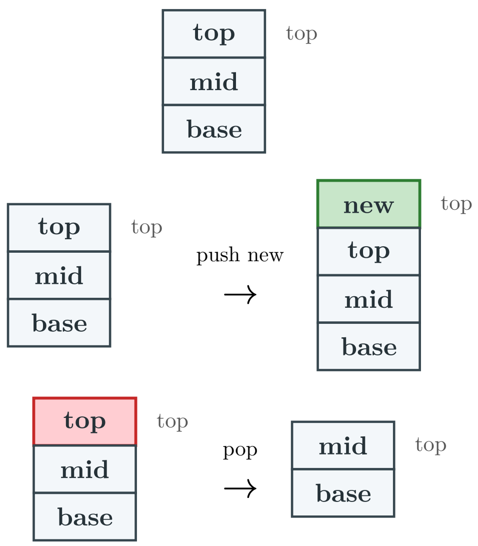


```typst
#stack(9, 7, 2, top-label:
[Peek]).diagram
```



Argument coverage: stacks use the linear `style` keys and operation marks for `.push` / `.pop` .

```typst
#let s = stack(
  [top], [mid], [base],
  style: (
    box-w: 1.25,
    box-h: 0.58,
    box-gap: 0,
    box-fill: rgb("#F3F7FA"),
    box-stroke: 0.8pt + rgb("#37474F"),
    node-text: (size: 9pt, color: rgb("#263238"), weight: "bold"),
    scale: 1.08,
    new-style: (fill: rgb("#C8E6C9"), stroke: 1pt + rgb("#2E7D32")),
    remove-style: (fill: rgb("#FFCDD2"), stroke: 1pt + rgb("#C62828")),
  ),
)
#std.stack(spacing: 1em, s.diagram, (s.push)([new]).diagram, (s.pop)().diagram)
```



3.6 `queue()`

Full signature:

```typst
#queue(
  ..vals,
  style: (:),
  enqueue: none,
  dequeue: none,
  front-label: [Front],
  rear-label: [Rear],
)
```

Argument	Type	Default	Description
`..vals`	`int` / `content`	(required)	Values in the queue; the **first** argument is the front.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`enqueue`	`none` / `int` / `content`	`none`	Draw one extra element entering at the rear, with an arrow, in the same frame.
`dequeue`	`none` / `int` / `content`	`none`	Draw the front element leaving to the left, with an arrow, in the same frame.
`front-label`	`content`	`[Front]`	Label over the front cell. Combined with `rear-label` for a one-cell queue.
`rear-label`	`content`	`[Rear]`	Label over the rear cell. Combined with `front-label` for a one-cell queue.

Nested keys for `style`

Path	Type	Default	Description
`style.box-w`	`float`	`0.95`	Cell width.
`style.box-h`	`float`	`0.7`	Cell height.
`style.box-gap`	`float`	`0.55`	Gap between cells or list nodes.
`style.box-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Cell outline.
`style.box-fill`	`color`	`white`	Default cell fill.
`style.ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `linked-list(pointer: true)`.
`style.prev-ptr-fill`	`color`	`rgb("#D7ECC9")`	Previous-pointer cell fill for `doubly-linked-list(pointer: true)`.
`style.next-ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `doubly-linked-list(pointer: true)`.
`style.node-text`	`dictionary`	`(size: 9pt)`	Cell text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Head, address, front, and rear label text dictionary. See nested keys below.

`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

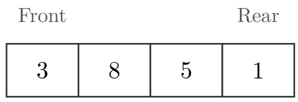
Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.

`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Field	Call	Effect
`.diagram`	n/a	The rendered queue.
`.enqueue`	`(obj.enqueue)(v)`	Enqueue `v` at the rear. Returns a step.
`.dequeue`	`(obj.dequeue)()`	Dequeue the front value (no argument). Returns a step.

Typst

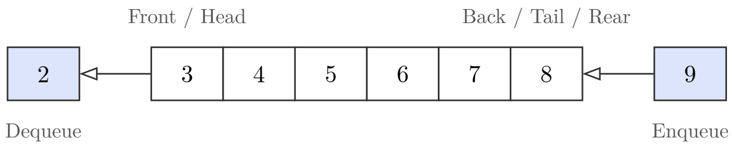
```
#queue(3, 8, 5, 1).diagram
```



`enqueue` / `dequeue` draw a single-frame arrow, independent of the object's own `.enqueue` / `.dequeue` methods:

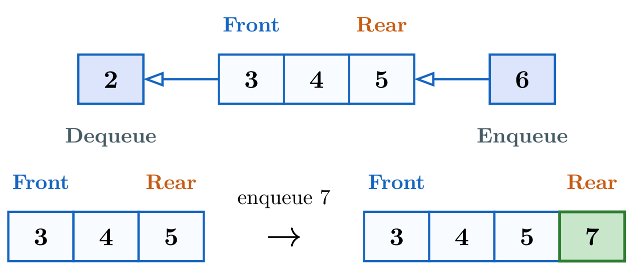
Typst

```
#queue(3, 4, 5, 6, 7, 8, enqueue: 9, dequeue: 2,
  front-label: [Front / Head], rear-label: [Back / Tail / Rear]).diagram
```



Argument coverage: queue examples can combine the single-frame `enqueue` / `dequeue` arrows, custom front/rear labels, label text styling, and operation mark styles.

```typst
#let q = queue(
  3, 4, 5,
  enqueue: 6,
  dequeue: 2,
  front-label: text(fill: rgb("#1565C0"))[*Front*],
  rear-label: text(fill: rgb("#C75B12"))[*Rear*],
  style: (
    box-w: 0.82,
    box-h: 0.62,
    box-gap: 0.35,
    box-fill: rgb("#F8FBFF"),
    box-stroke: 0.8pt + rgb("#1565C0"),
    node-text: (size: 9pt, weight: "bold"),
    label-text: (size: 7.5pt, color: rgb("#455A64"), weight: "bold"),
    scale: 1.05,
    new-style: (fill: rgb("#C8E6C9"), stroke: 1pt + rgb("#2E7D32")),
    remove-style: (fill: rgb("#FFCDD2"), stroke: 1pt + rgb("#C62828")),
  ),
)
#std.stack(spacing: 1em, q.diagram, (q.enqueue)(7).diagram)
```



3.7 `array-view()` and `matrix()`

Full signatures:

```typst
#array-view(..vals, style: (:), cell-customizations: (), pointers: ())
#matrix(rows, style: (:), cell-customizations: (), row-labels: none, column-labels: none)
```

These builders draw compact cell diagrams. They do not have operations yet; `cell-customizations` restyles individual array cells by index or matrix cells by `(row, column)` coordinate. `style.indices: (enabled: true)` draws zero-based array indices under the cells. `pointers` draws labelled arrows above array cells.

Argument	Type	Default	Description
`..vals`	`content`	(required for `array-view`)	Array cells, left to right.
`rows`	`array`	(required for `matrix`)	Rows of cells, written as nested arrays such as `((1, 2), (3, 4))` .

`cell-customizations`	`array / dictionary`	`()`	Cell overrides. For arrays, use `((index, options), ...)` . For matrices, use `(((row, col), options), ...)` .
`pointers`	`array`	`()`	`array-view` only. Labelled arrows above cells, each `(index: , label: , color: , text: )` . Markers sharing a cell spread across its width.
`row-labels`	`none / array`	`none`	Matrix only. Labels drawn to the left of rows.
`column-labels`	`none / array`	`none`	Matrix only. Labels drawn above columns.
`style`	`dictionary`	`(:)`	Cell style override. Uses `box-*` , `node-text` , `label-text` , and diff-highlight style keys.

Nested keys for `style`

Path	Type	Default	Description
`style.box-w`	`float`	`0.95`	Cell width.
`style.box-h`	`float`	`0.7`	Cell height.
`style.box-gap`	`float`	`0.55`	Gap between cells or list nodes.
`style.box-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Cell outline.
`style.box-fill`	`color`	`white`	Default cell fill.
`style.ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `linked-list(pointer: true)` .
`style.prev-ptr-fill`	`color`	`rgb("#D7ECC9")`	Previous-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.next-ptr-fill`	`color`	`rgb("#D7ECC9")`	Next-pointer cell fill for `doubly-linked-list(pointer: true)` .
`style.node-text`	`dictionary`	`(size: 9pt)`	Cell text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Head, address, front, and rear label text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.

`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Nested keys for `style.indices`

Path	Type	Default	Description
`style.indices.enabled`	`bool`	`false`	Draw index labels below array cells.

`style.indices.labels`	`auto` / `array`	`auto`	`auto` draws zero-based numeric indices; an array supplies custom labels.
`style.indices.offset`	`tuple`	`(0, -0.28)`	Canvas offset from the bottom-center index position.
`style.indices.size`	`length`	`style.label-text.size`	Index text size.
`style.indices.color`	`color`	`style.label-text.color`	Index text color.
`style.indices.font`	`str` / `array`	inherits	Typst text font selection.
`style.indices.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `cell-customizations`

Path	Type	Default	Description
`cell-customizations[]`	`tuple`	n/a	One `(cell, options)` tuple.
`cell-customizations[].cell`	`content`	required	Array index, or matrix coordinate `(row, column)` .
`cell-customizations[].options`	`dictionary`	required	Per-cell drawing options.
`cell-customizations[].options.fill`	`color`	normal fill	Cell fill.
`cell-customizations[].options.stroke`	`stroke`	normal stroke	Cell outline.
`cell-customizations[].options.text`	`dictionary`	`style.node-text`	Text style merged into the cell text.

Field	Call	Effect
`.diagram`	n/a	The rendered array or matrix.

Typst

```
#array-view(
  4, 1, 7, 3, 9,
  style: (
    indices: (
      enabled: true,
      color: rgb("#455A64"),
      weight: "bold",
    ),
  ),
  cell-customizations: (
    (1, (fill: rgb("#FFF3BF"), stroke: 1pt + rgb("#F59F00"))),
    (2, (fill: rgb("#D3F9D8"), stroke: 1pt + rgb("#2B8A3E"))),
  ),
).diagram

#matrix(
  ((0, 1, 0), (1, 0, 1), (0, 1, 0)),
  row-labels: ([A], [B], [C]),
  column-labels: ([A], [B], [C]),
  cell-customizations: (
    ((0, 1), (fill: rgb("#E7F5FF"), stroke: 1pt + rgb("#1971C2"))),
    ((1, 2), (fill: rgb("#D3F9D8"), stroke: 1pt + rgb("#2B8A3E"))),
  ),
).diagram
```

4	1	7	3	9
0	**1**	**2**	**3**	**4**

	A	B	C
A	0	1	0
B	1	0	1
C	0	1	0

3.8 Sorting algorithms

Full signatures:

```typst
#merge-sort(array, order: "asc", labels: true)
#merge-operation(left, right, order: "asc", pointers: true, labels: true)
#partition-step(array, order: "asc", pivot: "middle", pointers: false, labels: true)
#quick-sort(array, order: "asc", pivot: "last", labels: true)
#bubble-sort(array, order: "asc", pointers: true, labels: true, compare: none, swap: none)
#insertion-sort(array, order: "asc", pointers: true, labels: true, compare: none, swap: none)
#selection-sort(array, order: "asc", pointers: true, labels: true,
  compare: none, current: none, minimum: none, swap: none)
#sort-sequence(steps, columns: 3, gap: 1em, row-gap: 1em)
```

Sorting builders generate full traces from a single input array. Each returns `.diagram` , `.steps` , and `.result` . Use `.diagram` for the default wrapped trace, or pass `.steps` to `sort-sequence(...)` when you want different columns or spacing. Every trace records all comparison and swap steps. Each step keeps its label and array together when a page break is needed. Merge sort renders a connected tree: arrows lead from each array to its two divided arrays, then from each pair into its merged result. Curly braces identify the Divide, Merge, and Partition phases. Quicksort groups all active recursive partitions on the same row before expanding them together on the next row. Use `partition-step(..., pivot: "last")` for the detailed partition used by a last-pivot quicksort; `pointers: true` adds labelled cursor arrows, and each successful comparison includes the subsequent `i` advance as its own frame.

To style a trace, pass a styled `array-view(...)` in place of the bare array: its style (fills, text, and index configuration) is carried through every step of the trace. Bubble sort adds a pointer-free settled frame after each pass to highlight its green suffix, and selection sort adds one after each minimum swap to highlight its sorted prefix. Settled frames name the newly settled value, and selection scan labels list position, minimum, then item.

Merge operation, bubble, insertion, and selection sort show labelled arrows above their active positions by default. Pass `pointers: false` to hide them. Merge operation labels its left, right, and output cursors `i` , `j` , and `i+j` respectively. Each arrow sits over a marked cell, and several markers on one cell spread across its width. The arrows are additive: the role still fills its cells, so pointers layer on top of the fill. Each role (`compare` , `swap` , and, for selection sort, `current` and `minimum`) takes a `node-mark-style(...)` that restyles that role's fill, stroke, and text in either mode; the arrow colour comes from the mark's stroke, or its fill when no stroke is set. To keep a cell unfilled under a pointer, set the role's `fill` to `none` . Quick sort and merge sort render composite phase diagrams and do not take marker roles.

Pass `labels: false` to hide generated text captions while preserving the arrays, highlights, indices, and optional pointer arrows. This also hides merge's left/right/result headings, partition metadata, and composite phase labels, which lets you supply your own translated explanation alongside a trace.

Sorting traces work directly with Touying reveals: `.steps` is an ordered array of rendered frames, so it can be revealed one at a time.

```typst
#let trace = bubble-sort((5, 1, 4, 2))

#for (index, step) in trace.steps.enumerate() [
  #only(index + 1)[#step.diagram]
]
```

`only(index + 1)` shows exactly one step per reveal in the same spot. Using `#pause` instead accumulates the steps, stacking every frame on screen at once.

Argument	Type	Default	Description
`array`	`array` / `array-view`	(required)	Values to sort from left to right, as a plain array or a styled `array-view(...)` whose style is reused for every step. Values must be comparable with `<` / `>`. `merge-operation` takes two such arguments (`left`, `right`); every other builder takes one.
`order`	`"asc"` / `"desc"`	`"asc"`	Sort direction.
`pivot`	`"first"` / `"last"` / `int` / `"middle"`	`"last"` / `"middle"`	Quicksort selects an end of each subarray or a position index. `partition-step` accepts `middle` or `last` use `last` for the detailed quicksort partition trace.
`pointers`	`bool`	`true` / `false`	Merge operation, bubble, insertion, selection sort, and last-pivot `partition-step` only. Draw labelled index arrows above the marked cells, layered on top of their fills. Merge operation uses `i`, `j`, and `i+j` last-pivot partition uses `i`, `j`, and `pivot`.
`labels`	`bool`	`true`	All sorting builders. Set to `false` to hide generated step captions and structural text while preserving diagrams, highlights, indices, and pointer arrows.
`compare`, `swap`, `current`, `minimum`	`dictionary`	`none`	Bubble, insertion, and selection sort only. A `node-mark-style(...)` override for that role's marker. `current` and `minimum` apply to selection sort only.

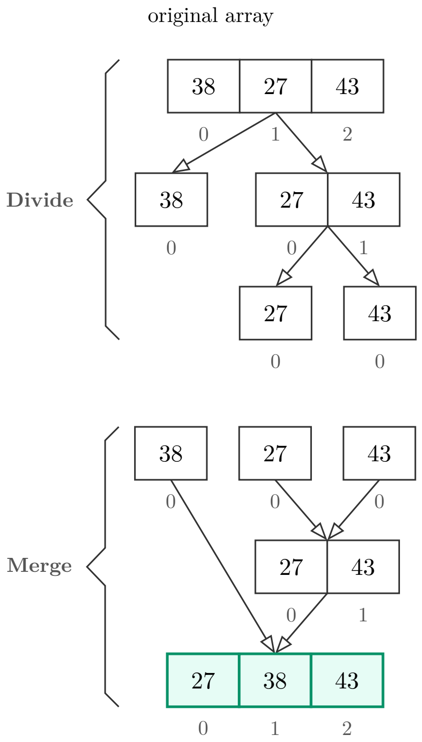
Field	Call	Effect
`.diagram`	n/a	Rendered full sorting trace.
`.steps`	`array`	Generated step dictionaries. Each step has `label`, `diagram`, and `values`.
`.result`	`array`	Sorted values.

Full traces

Long traces below flow one labelled step at a time, so page breaks cannot separate a label from its array or make neighbouring frames overlap.

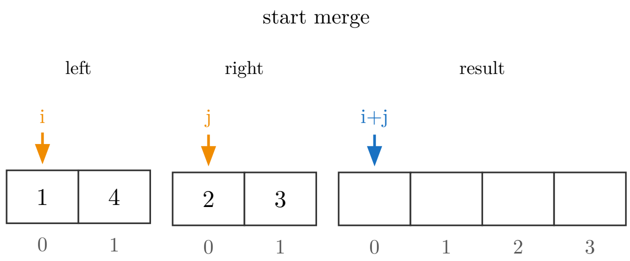
Merge sort

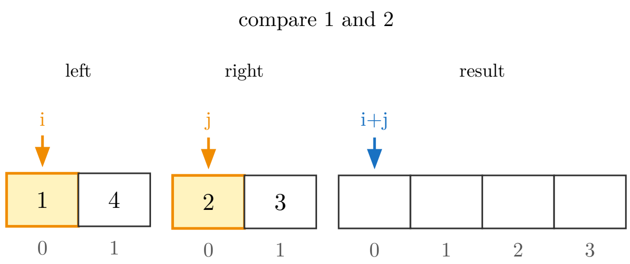
```typst
#let trace = merge-sort((38, 27, 43))
#trace.diagram
```

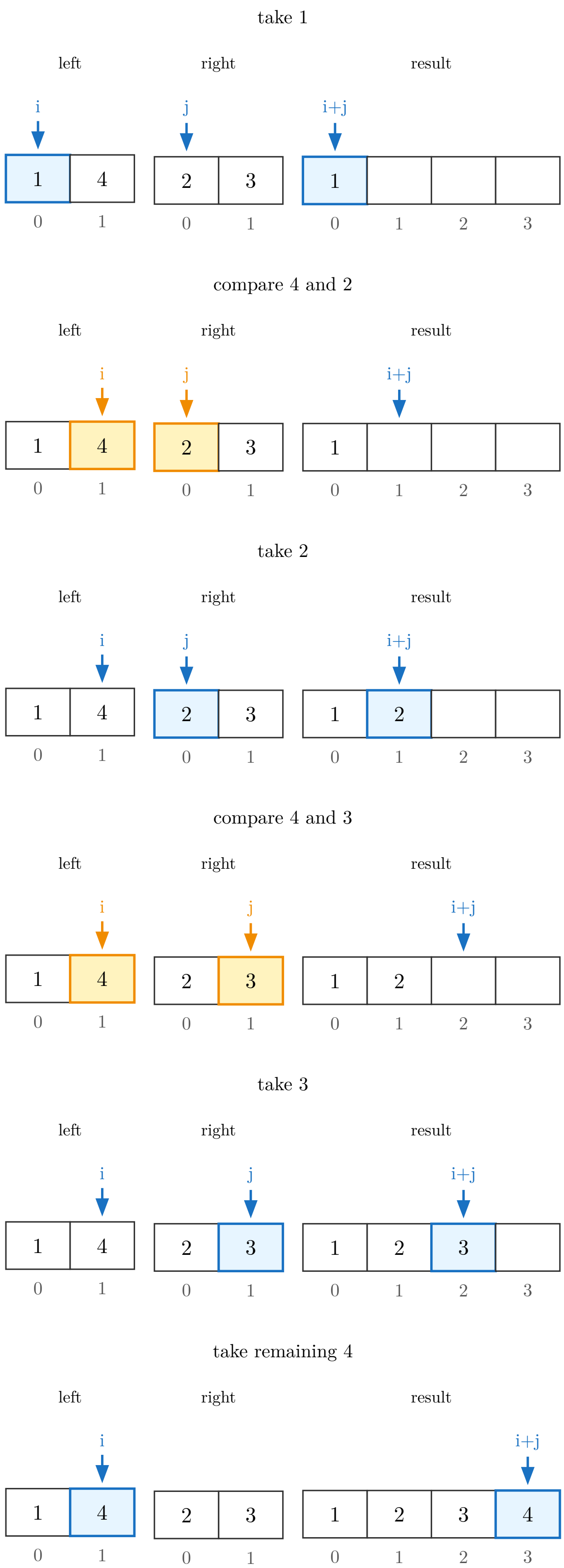


Merge operation

```typst
#let trace = merge-operation((1, 4), (2, 3))
#for step in trace.steps [#step.diagram]
```

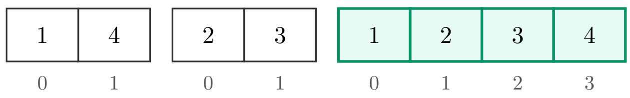





take 1
left
right
result
i
j
i+j
1 4
2 3
1
0 1
0 1
0 1 2 3
compare 4 and 2
left
right
result
i
j
i+j
1 4
2 3
1
0 1
0 1
0 1 2 3
take 2
left
right
result
i
j
i+j
1 4
2 3
1 2
0 1
0 1
0 1 2 3
compare 4 and 3
left
right
result
i
j
i+j
1 4
2 3
1 2
0 1
0 1
0 1 2 3
take 3
left
right
result
i
j
i+j
1 4
2 3
1 2 3
0 1
0 1
0 1 2 3
take remaining 4
left
right
result
i
i+j
1 4
2 3
1 2 3 4
0 1
0 1
0 1 2 3

merged

left right result

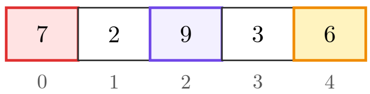


Partition around a middle pivot

```typst
#let trace = partition-step((7, 2, 9, 3, 6))
#for step in trace.steps [#step.diagram]
```

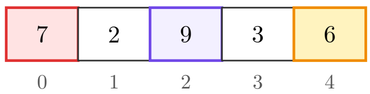
start

pivot: 9 at 2 i: 0 j: 4



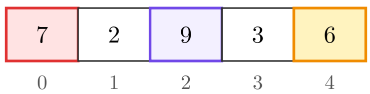
select pivot 9

pivot: 9 at 2 i: 0 j: 4



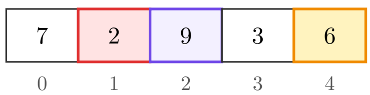
i 7 satisfies the pivot test

pivot: 9 at 2 i: 0 j: 4



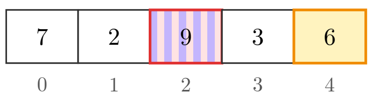
i 2 satisfies the pivot test

pivot: 9 at 2 i: 1 j: 4



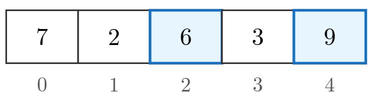
compare 9 and 6

pivot: 9 at 2 i: 2 j: 4



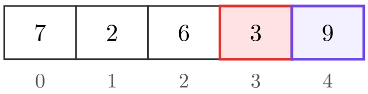
swap 9 and 6

pivot: 9 at 4 i: 2 j: 4



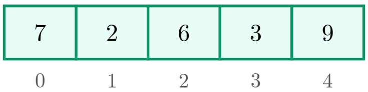
i 3 satisfies the pivot test

pivot: 9 at 4 i: 3 j: 3



partitioned

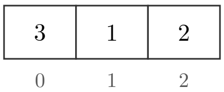
pivot: 9 at 4 i: 4 j: 3

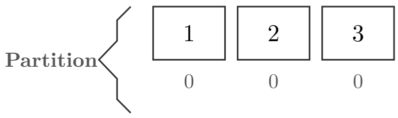


Quick sort

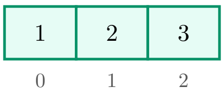
```typst
#let trace = quick-sort((3, 1, 2))
#trace.diagram
```

original array





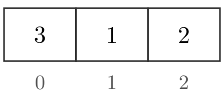
sorted array



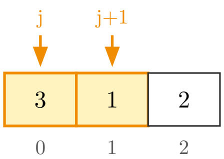
Bubble sort

```typst
#let trace = bubble-sort((3, 1, 2))
#for step in trace.steps [#step.diagram]
```

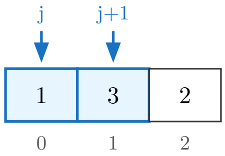
start



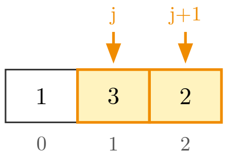
compare 3 and 1



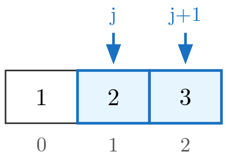
swap 3 and 1



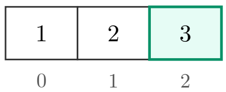
compare 3 and 2



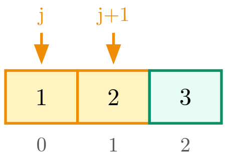
swap 3 and 2



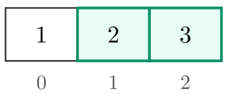
3 settled



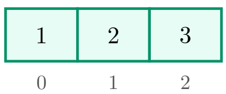
compare 1 and 2



2 settled

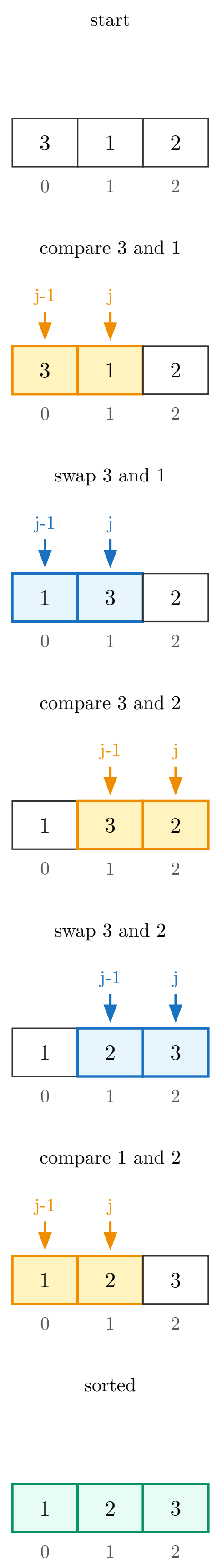


sorted



Insertion sort

```typst
#let trace = insertion-sort((3, 1, 2))
#for step in trace.steps [#step.diagram]
```

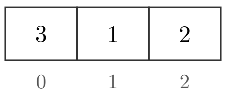


Selection sort

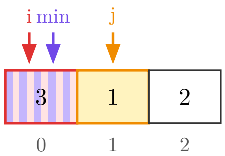
```typst
#let trace = selection-sort((3, 1, 2))
#for step in trace.steps [#step.diagram]
```

Typst

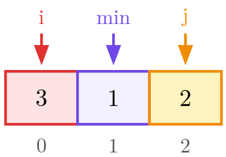
start



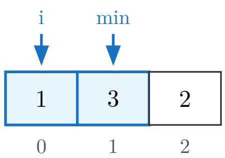
position 0, minimum 0, item 1



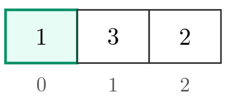
position 0, minimum 1, item 2



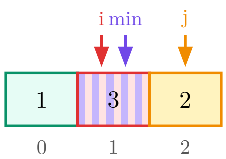
swap 3 and 1



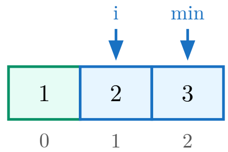
1 settled



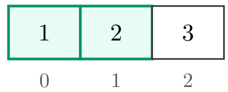
position 1, minimum 1, item 2



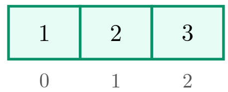
swap 3 and 2



2 settled

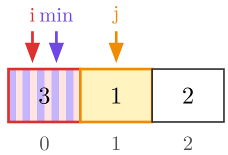


sorted



Hide generated captions

```typst
#let trace = selection-sort((3, 1, 2), labels: false)
#trace.steps.at(1).diagram
```

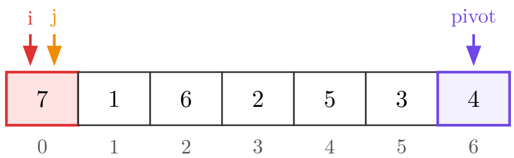


With `pivot: "last"` , placing the median at the end produces an even quicksort partition: three values on each side of the pivot. This trace renders every comparison, swap, and cursor position.

```typst
#let trace = partition-step(
  (7, 1, 6, 2, 5, 3, 4),
  pivot: "last",
  pointers: true,
)
#for step in trace.steps [#step.diagram]
```

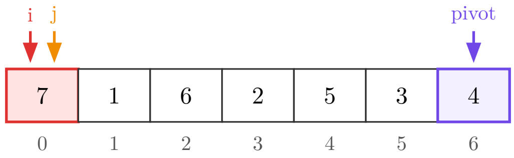
start

pivot: 4 at 6 i: 0 j: 0



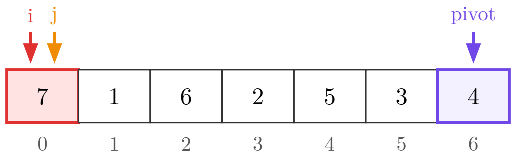
select last pivot 4

pivot: 4 at 6 i: 0 j: 0



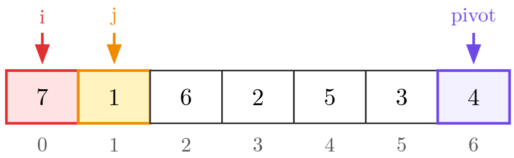
compare 7 and pivot 4

pivot: 4 at 6 i: 0 j: 0



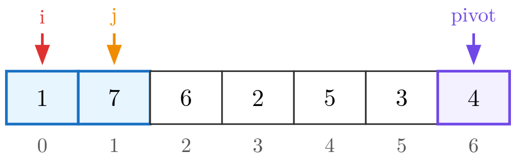
compare 1 and pivot 4

pivot: 4 at 6 i: 0 j: 1



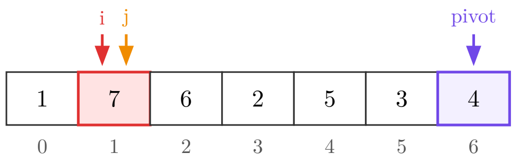
swap 7 and 1

pivot: 4 at 6 i: 0 j: 1



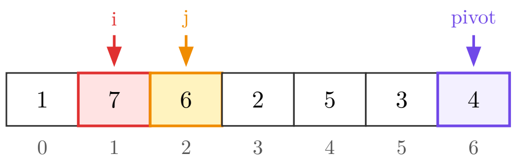
advance i to 1

pivot: 4 at 6 i: 1 j: 1



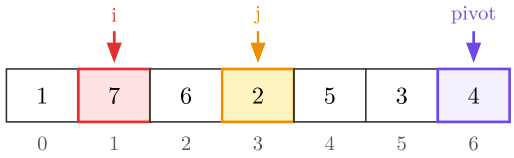
compare 6 and pivot 4

pivot: 4 at 6 i: 1 j: 2



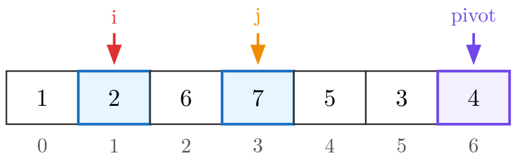
compare 2 and pivot 4

pivot: 4 at 6 i: 1 j: 3



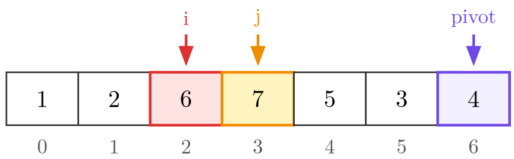
swap 7 and 2

pivot: 4 at 6 i: 1 j: 3



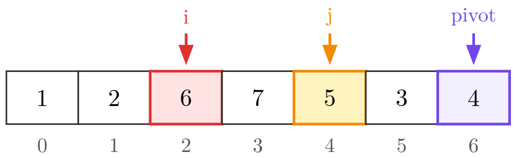
advance i to 2

pivot: 4 at 6 i: 2 j: 3



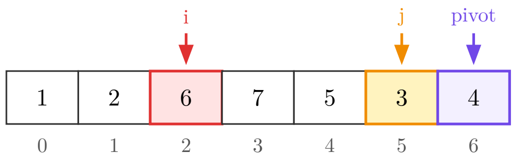
compare 5 and pivot 4

pivot: 4 at 6 i: 2 j: 4



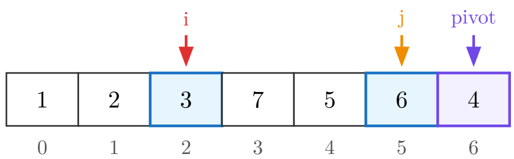
compare 3 and pivot 4

pivot: 4 at 6 i: 2 j: 5



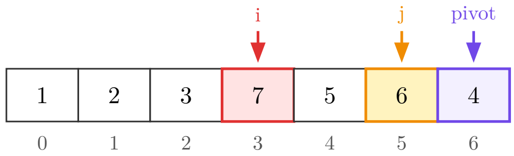
swap 6 and 3

pivot: 4 at 6 i: 2 j: 5



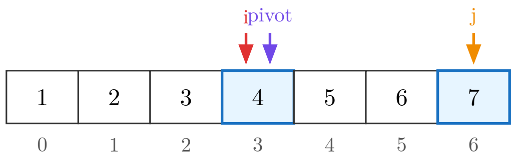
advance i to 3

pivot: 4 at 6 i: 3 j: 5



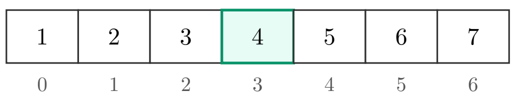
place pivot 4

pivot: 4 at 3 i: 3 j: 6



partitioned

pivot: 4 at 3 i: 3 j: 6

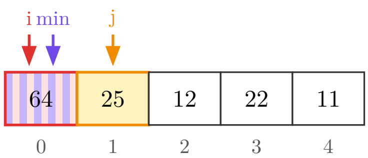


Pointer markers keep the cell fill clean and let each role carry its own colour. These focused frames show selection's `i` , `min` , and `j` markers together, plus customized bubble-sort marker strokes.

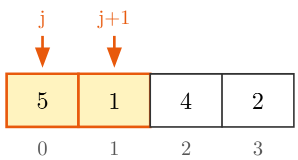
```typst
#let selection = selection-sort((64, 25, 12, 22, 11), pointers: true)
#selection.steps.at(1).diagram

#let bubble = bubble-sort(
  (5, 1, 4, 2),
  pointers: true,
  compare: node-mark-style(stroke: 1pt + rgb("#E8590C")),
  swap: node-mark-style(stroke: 1pt + rgb("#2F9E44")),
)
#bubble.steps.at(1).diagram
```

position 0, minimum 0, item 1



compare 5 and 1



3.9 `graph()`

Full signature:

```typst
#graph(
  adjacency,
  directed: true,
  labels: (:),
  positions: (:),
  layout: "auto",
  radius: auto,
  edge-customizations: (),
  node-customizations: (),
  node-labels: (:),
  style: (:),
)
```

`adjacency` maps each node label to an array of that node's neighbor labels. To label an edge, use `(neighbor, label)` instead of just `neighbor` . A node with no outgoing edges still needs its key present, with an empty array; a label that only ever shows up as someone else's neighbor gets added on its own.

Argument	Type	Default	Description
`adjacency`	`dictionary`	(required)	Node label `(str)` to array of neighbor labels or `(neighbor, edge-label)` pairs.
`directed`	`bool`	`true`	Draw an arrowhead on every declared pair. `false` drops arrowheads and collapses a reciprocal pair into the one edge it represents.
`labels`	`dictionary`	`(:)`	Node label to drawn content: math, styled text, an image, anything. A label left out keeps the plain key as its own content.
`positions`	`dictionary`	`(:)`	Node label to absolute `(x, y)` or `(rel:, offset:)` placement.
`layout`	`str`	`"auto"`	`"auto"` starts from the circular layout and applies `positions` . `"manual"` requires every node in `positions` .
`radius`	`auto` / `float`	`auto`	Circle radius for `layout: "auto"` . Passing a radius with `layout: "manual"` is an error.
`edge-customizations`	`array`	`()`	`(from, to, options)` tuples restyling one edge. See below.
`node-customizations`	`array`	`()`	`(node, options)` tuples restyling one node by graph identity.
`node-labels`	`array` / `dictionary`	`(:)`	Labels drawn outside individual graph nodes.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.

Nested keys for `adjacency`

Path	Type	Default	Description
adjacency.<node>	array	required	Outgoing neighbors for one node. Use an empty array to draw an isolated node.
adjacency.<node>[]	content / tuple	required	Either a neighbor label, or (neighbor, edge-label) to draw a label on that edge.
adjacency.<node>[].neighbor	content	required	Target node identity.
adjacency.<node>[].edge-label	content	none	Optional edge label, such as a weight.

Nested keys for labels

Path	Type	Default	Description
labels.<node>	content	node key	Drawn content for one node. Identity still comes from the plain adjacency/position label.

Nested keys for positions

Path	Type	Default	Description
positions.<node>	point / dictionary	auto layout position	Absolute (x, y) position or relative placement dictionary.
positions.<node>.rel	content	required for relative placement	Another node label to place from.
positions.<node>.offset	point	required for relative placement	(dx, dy) offset from rel .

Nested keys for edge-customizations

Path	Type	Default	Description
edge-customizations[]	tuple	n/a	One (from, to, options) tuple. For trees, from / to are parent and child values; for graphs, node labels.
edge-customizations[].from	content	required	Source node identity.
edge-customizations[].to	content	required	Target node identity.
edge-customizations[].options	dictionary	required	Per-edge options.
edge-customizations[].options.stroke	stroke	none	Full stroke override. Wins over color if both are set.
edge-customizations[].options.color	color	none	Paint only, at the default thickness.
edge-customizations[].options.pattern	str	"normal"	"normal" , "dashed" , "dotted" , or "wavy" .
edge-customizations[].options.arrow	none / bool / str	none	Per-edge arrowheads: none , "end" / true , "start" , or "both" .

`edge-customizations[].options.bend`	`str` / `bool`	`false`	`"left"` or `"right"` curves the edge relative to its travel direction. `false` keeps it straight.
`edge-customizations[].options.angle`	`angle`	`25deg`	How sharply a bent edge curves. Ignored when `bend` is `false` .
`edge-customizations[].options.label`	`content` / `dictionary`	none	Tree edge label content, or graph edge-label text overrides. A dictionary can include `content` for trees.

Nested keys for `edge-customizations[].options.label`

Path	Type	Default	Description
`edge-customizations[].options.label.size`	`length`	`style.label-text.size`	Edge-label size.
`edge-customizations[].options.label.content`	`content`	tree only	Tree edge label content. Graph edge labels come from adjacency entries such as `("B", [4])` .
`edge-customizations[].options.label.color`	`color`	`style.label-text.color`	Edge-label text color.
`edge-customizations[].options.label.rotation`	`angle` / `str`	`style.label-text.rotation`	An angle, or `"edge"` to follow the tree/graph edge angle.
`edge-customizations[].options.label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`edge-customizations[].options.label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `style`

Path	Type	Default	Description
`style.node-radius`	`float`	`0.34`	Circle radius, or half the square/diamond/hexagon size, in canvas units.
`style.node-shape`	`str`	`"circle"`	`"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.node-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Node outline.
`style.node-fill`	`color`	`white`	Default node fill.
`style.edge-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Default edge stroke.
`style.edge-arrow`	`none` / `bool` / `str`	`none`	Global arrowhead override: `none` / `false` , `"end"` or `true` , `"start"` , or `"both"` . `directed: true` uses `"end"` unless this overrides it.
`style.edge-arrow-fill`	`none` / `color`	`none`	`none` gives open arrowheads; a color fills arrowheads solid.
`style.edge-pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`style.edge-wave-amplitude`	`float`	`0.07`	Wave height in canvas units.

`style.edge-wave-step`	`float`	`0.14`	Approximate wavelength in canvas units.
`style.node-text`	`dictionary`	`(size: 9pt)`	Node text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Edge label text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on the rendered graph.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `node-customizations`

Path	Type	Default	Description
`node-customizations[]`	`tuple`	n/a	One `(node, options)` tuple.
`node-customizations[].node`	`content`	required	Graph identity key, not the display label from `labels`.
`node-customizations[].options`	`dictionary`	required	Per-node drawing options.
`node-customizations[].options.fill`	`color`	normal fill	Node fill.
`node-customizations[].options.stroke`	`stroke`	normal stroke	Node outline.
`node-customizations[].options.shape`	`str`	`style.node-shape`	Trees/graphs only: `"circle"`, `"square"`, `"diamond"`, or `"hexagon"`.
`node-customizations[].options.node-radius`	`float`	`style.node-radius`	Trees/graphs only. Shape size.

`node-customizations[].options.text`	`dictionary`	`style.node-text`	Text style merged into the node text.

Nested keys for `node-labels`

Path	Type	Default	Description
`node-labels[]`	`tuple`	n/a	One `(node, label)` tuple.
`node-labels[].node`	`content`	required	Graph identity key, not the display label from `labels` .
`node-labels[].label`	`content` / `dictionary`	required	Bare content for quick labels, or a dictionary with `content` plus style and placement keys.
`node-labels[].label.content`	`content`	required for dict labels	Label body. `body` is also accepted.
`node-labels[].label.position`	`str` / `angle`	`"right"`	`"right"` , `"left"` , `"top"` , `"bottom"` , or an angle.
`node-labels[].label.offset`	`point`	`(0, 0)`	Extra `(dx, dy)` shift after the position is applied.
`node-labels[].label.size`	`length`	`style.label-text.size`	Node-label size.
`node-labels[].label.color`	`color`	`style.label-text.color`	Node-label text color.
`node-labels[].label.rotation`	`angle`	`style.label-text.rotation`	Rotation of the label text itself.
`node-labels[].label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`node-labels[].label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

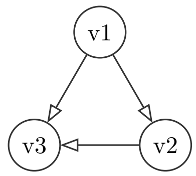
Nested keys for `style.node-labels`

Path	Type	Default	Description
`style.node-labels.position`	`str` / `angle`	`"right"`	Default position for all node labels in this diagram.
`style.node-labels.offset`	`point`	`(0, 0)`	Default extra `(dx, dy)` shift for all node labels.
`style.node-labels.gap`	`float`	`0.22`	Default distance from the node boundary in canvas units.
`style.node-labels.size`	`length`	`style.label-text.size`	Default node-label text size.
`style.node-labels.color`	`color`	`style.label-text.color`	Default node-label text color.
`style.node-labels.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-labels.weight`	`str` / `int`	inherits	Typst text weight.
`style.node-labels.rotation`	`angle`	`style.label-text.rotation`	Default rotation of the label text itself.

Field	Call	Effect

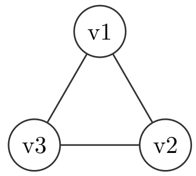
`.diagram`	n/a	The rendered graph.

```typst
#graph(("v1": ("v2", "v3"), "v2":
("v3",), "v3": ())).diagram
```



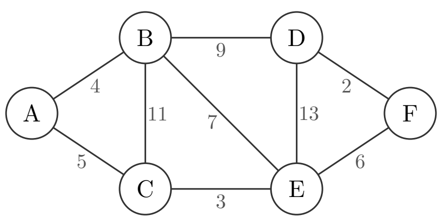
Declaring both directions of an edge still draws it once.

```typst
#graph(
  ("v1": ("v2", "v3"), "v2": ("v1", "v3"),
  "v3": ()),
  directed: false,
).diagram
```



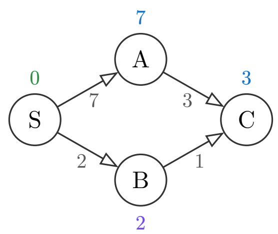
Use `(neighbor, label)` entries to draw edge labels, such as weights.

```typst
#graph(
  (
    "A": (("B", [4]), ("C", [5])),
    "B": (("C", [11]), ("D", [9]), ("E", [7])),
    "C": (("E", [3]),),
    "D": (("E", [13]), ("F", [2])),
    "E": (("F", [6]),),
    "F": (),
  ),
  directed: false,
  layout: "manual",
  positions: (
    "A": (0, 0),
    "B": (rel: "A", offset: (1.5, 1.0)),
    "C": (rel: "A", offset: (1.5, -1.0)),
    "D": (rel: "B", offset: (2.0, 0.0)),
    "E": (rel: "C", offset: (2.0, 0.0)),
    "F": (rel: "D", offset: (1.5, -1.0)),
  ),
).diagram
```



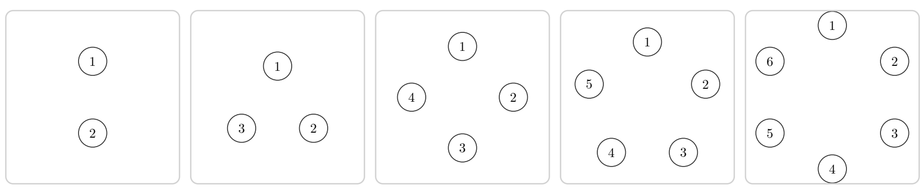
Use `node-labels` for annotations such as tentative distances, predecessor tags, ranks, or algorithm state. The node's identity and its drawn inside label stay unchanged.

```typst
#graph(
  ("S": (("A", [7]), ("B", [2])), "A": (("C", [3]),), "B": (("C", [1]),), "C": ()),
  layout: "manual",
  positions: ("S": (0, 0), "A": (1.4, 0.8), "B": (1.4, -0.8), "C": (2.8, 0)),
  node-labels: (
    ("S", (content: [$0$], position: "top", color: rgb("#2B8A3E"))),
    ("A", [$7$]),
    ("B", (content: [$2$], position: "bottom", color: rgb("#7048E8"))),
    ("C", [$3$]),
  ),
  style: (
    edge-arrow: "end",
    node-labels: (position: "top", color: rgb("#1971C2")),
  ),
).diagram
```



`layout: "auto"` is the default. Nodes are placed around a circle; the radius grows as the node count increases. Pass `radius:` to set the circle radius yourself.

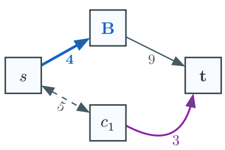
```typst
#let panel(body) = block(width: 6.2em, height: 6.2em, inset: 0.2em,
  stroke: 0.5pt + luma(210), radius: 3pt, clip: true,
  align(center + horizon, scale(55%, reflow: true, body)))
#std.stack(dir: ltr, spacing: 0.4em,
  panel(graph(("1": (), "2": ())).diagram),
  panel(graph(("1": (), "2": (), "3": ())).diagram),
  panel(graph(("1": (), "2": (), "3": (), "4": ())).diagram),
  panel(graph(("1": (), "2": (), "3": (), "4": (), "5": ())).diagram),
  panel(graph(("1": (), "2": (), "3": (), "4": (), "5": (), "6": ())).diagram),
)
```



`radius` only applies to `layout: "auto"` . In manual layout, every node already has an explicit position, so combining `radius` with `layout: "manual"` raises an error.

Argument coverage: this graph combines adjacency edge labels, node display labels, manual and relative positions, graph-wide style, per-edge options, and nested edge-label overrides.

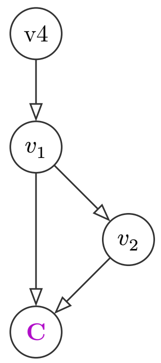
```typst
#graph(
  (
    "A": (("B", [4]), ("C", [5])),
    "B": (("D", [9]),),
    "C": (("D", [3]),),
    "D": (),
  ),
  directed: true,
  labels: (
    "A": $s$,
    "B": text(fill: rgb("#1565C0"))[*B*],
    "C": $c_1$,
    "D": text(weight: "bold")[t],
  ),
  layout: "manual",
  positions: (
    "A": (0, 0),
    "B": (rel: "A", offset: (1.6, 0.9)),
    "C": (rel: "A", offset: (1.6, -0.9)),
    "D": (rel: "B", offset: (1.8, -0.9)),
  ),
  edge-customizations: (
    ("A", "B", (stroke: 1.4pt + rgb("#1565C0"), label: (color: rgb("#1565C0"), weight: "bold"))),
    ("A", "C", (pattern: "dashed", arrow: "both", label: (rotation: "edge"))),
    ("C", "D", (bend: "right", angle: 35deg, color: rgb("#8E24AA"), label: (size: 8pt, color: rgb("#8E24AA")))),
  ),
  style: (
    node-shape: "square",
    node-radius: 0.34,
    node-fill: rgb("#F8FBFF"),
    node-stroke: 0.8pt + rgb("#37474F"),
    node-text: (size: 9pt, color: rgb("#263238")),
    edge-stroke: 0.7pt + rgb("#455A64"),
    edge-arrow: "end",
    edge-arrow-fill: rgb("#455A64"),
    edge-pattern: "normal",
    label-text: (size: 7.5pt, color: rgb("#555555")),
    scale: 0.7,
  ),
).diagram
```



3.10 `labels` and `positions`

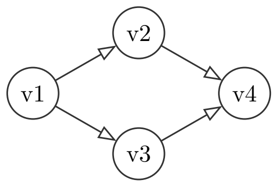
`labels` swaps what's drawn inside a node for any content, keyed by the plain-string label that `adjacency` / `positions` / `edge-customizations` still use for identity. With the default `layout: "auto"` , `positions` places a node at an absolute `(x, y)` point or at `(rel: other-label, offset: (dx, dy))` on top of the automatic circular layout.

Typst
```
#graph(
  ("v1": ("v2", "v3"), "v2": ("v3",), "v3": (), "v4": ("v1",)),
  labels: ("v1": $v_1$, "v2": $v_2$, "v3": text(fill: purple)[*C*]),
  positions: ("v4": (rel: "v1", offset: (0, 1.5))),
).diagram
```



Set `layout: "manual"` to skip the circular layout. Every node, including a neighbor-only node, must have a `positions` entry. At least one absolute `(x, y)` entry should act as an anchor for relative placements.

Typst
```
#graph(
  ("v1": ("v2", "v3"), "v2": ("v4",), "v3": ("v4",), "v4": ()),
  layout: "manual",
  positions: (
    "v1": (0, 0),
    "v2": (rel: "v1", offset: (1.4, 0.8)),
    "v3": (rel: "v1", offset: (1.4, -0.8)),
    "v4": (rel: "v2", offset: (1.4, -0.8)),
  ),
).diagram
```



3.11 `edge-customizations`

An array of `(from, to, options)` tuples, each restyling one edge. For `graph` , `from` and `to` are node labels. For generated trees, they are parent and child values. `options` is a dictionary.

Nested keys for `edge-customizations`

Path	Type	Default	Description
`edge-customizations[]`	`tuple`	n/a	One `(from, to, options)` tuple. For trees, `from` / `to` are parent and child values; for graphs, node labels.
`edge-customizations[].from`	`content`	required	Source node identity.
`edge-customizations[].to`	`content`	required	Target node identity.
`edge-customizations[].options`	`dictionary`	required	Per-edge options.
`edge-customizations[].options.stroke`	`stroke`	none	Full stroke override. Wins over `color` if both are set.
`edge-customizations[].options.color`	`color`	none	Paint only, at the default thickness.
`edge-customizations[].options.pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`edge-customizations[].options.arrow`	`none` / `bool` / `str`	none	Per-edge arrowheads: `none` , `"end"` / `true` , `"start"` , or `"both"` .
`edge-customizations[].options.bend`	`str` / `bool`	`false`	`"left"` or `"right"` curves the edge relative to its travel direction. `false` keeps it straight.
`edge-customizations[].options.angle`	`angle`	`25deg`	How sharply a bent edge curves. Ignored when `bend` is `false` .
`edge-customizations[].options.label`	`content` / `dictionary`	none	Tree edge label content, or graph edge-label text overrides. A dictionary can include `content` for trees.

Nested keys for `edge-customizations[].options.label`

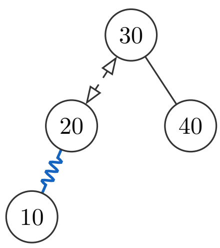
Path	Type	Default	Description
`edge-customizations[].options.label.size`	`length`	`style.label-text.size`	Edge-label size.
`edge-customizations[].options.label.content`	`content`	tree only	Tree edge label content. Graph edge labels come from adjacency entries such as `("B", [4])` .
`edge-customizations[].options.label.color`	`color`	`style.label-text.color`	Edge-label text color.
`edge-customizations[].options.label.rotation`	`angle` / `str`	`style.label-text.rotation`	An angle, or `"edge"` to follow the tree/graph edge angle.
`edge-customizations[].options.label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`edge-customizations[].options.label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Typst

```
#graph(
  ("v1": ("v2", "v3"), "v2": ("v3",), "v3": ()),
  edge-customizations: (
    ("v1", "v2", (stroke: 2pt + red)),
    ("v2", "v3", (color: rgb("#2E7D32"))),
    ("v1", "v3", (pattern: "wavy", arrow: "both")),
  ),
```

Tree edge customizations use the values in the parent and child nodes:

```
#bst(
  30, 20, 40, 10,
  edge-customizations: (
    (30, 20, (pattern: "dashed", arrow: "both")),
    (20, 10, (pattern: "wavy", color: rgb("#1565C0"))),
  ),
).diagram
```



Note. A mutual pair bent the same way on both directions (`("v1", "v2", (bend: "left"))` and `("v2", "v1", (bend: "left"))`) curves to opposite sides, since `"left"` is relative to each edge's own direction of travel. The two arrows read as two edges instead of one double-headed line.

Note. Nodes default to a circle, in the order their labels first show up, radius growing with the node count. Connected nodes don't cluster closer together yet, and `graph` has no operations: no add-node, add-edge, or a transition to match trees and heaps. See Section 9.

4 Hand-composed trees

For abstract diagrams, build a tree by hand with `node(...)` and `subtree(...)` instead of inserting real keys. Draw a node over two elided subtrees, the way you'd draw an AVL height invariant.

```typst
#tree(root, style: (:), edge-customizations: (), node-customizations: (), node-labels: (:))
#node(label, left: none, right: none, children: none, fill: none)
#subtree(label, fill: none, height: none, scale: 1)
```

Argument	Type	Default	Description
`root`	result of `node()`	(required)	The tree to render.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`edge-customizations`	`array`	`()`	`(parent, child, options)` tuples restyling individual tree edges by node label. See Section 3.11.
`node-customizations`	`array`	`()`	`(node, options)` tuples restyling individual nodes by label.
`node-labels`	`array` / `dictionary`	`(:)`	Labels drawn outside individual nodes.

Nested keys for `style`

Path	Type	Default	Description
`style.node-radius`	`float`	`0.34`	Circle radius, or half the square/diamond/hexagon size, in canvas units.
`style.node-shape`	`str`	`"circle"`	`"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.x-gap`	`float`	`1.05`	Horizontal spacing between in-order columns.
`style.y-gap`	`float`	`1.2`	Vertical spacing between depths.
`style.node-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Node outline.
`style.node-fill`	`color`	`white`	Default node fill.
`style.edge-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Parent-child edge stroke.
`style.edge-arrow`	`none` / `bool` / `str`	`none`	Edge arrowheads: `none` / `false` , `"end"` or `true` , `"start"` , or `"both"` .
`style.edge-pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`style.edge-wave-amplitude`	`float`	`0.07`	Wave height in canvas units.
`style.edge-wave-step`	`float`	`0.14`	Approximate wavelength in canvas units.

`style.node-text`	`dictionary`	`(size: 9pt)`	Node text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Annotation text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested subtree style keys

Path	Type	Default	Description
`style.tri-w`	`float`	`1.2`	Triangle base width before `subtree(scale:)` .
`style.tri-h`	`float`	`1.4`	Triangle height before `subtree(scale:)` .

Nested keys for diff-highlight dictionaries

Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.

`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.
`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Nested keys for `edge-customizations`

Path	Type	Default	Description
`edge-customizations[]`	`tuple`	n/a	One `(from, to, options)` tuple. For trees, `from` / `to` are parent and child values; for graphs, node labels.
`edge-customizations[].from`	`content`	required	Source node identity.
`edge-customizations[].to`	`content`	required	Target node identity.
`edge-customizations[].options`	`dictionary`	required	Per-edge options.
`edge-customizations[].options.stroke`	`stroke`	none	Full stroke override. Wins over `color` if both are set.
`edge-customizations[].options.color`	`color`	none	Paint only, at the default thickness.
`edge-customizations[].options.pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`edge-customizations[].options.arrow`	`none` / `bool` / `str`	none	Per-edge arrowheads: `none` , `"end"` / `true` , `"start"` , or `"both"` .
`edge-customizations[].options.bend`	`str` / `bool`	`false`	`"left"` or `"right"` curves the edge relative to its travel direction. `false` keeps it straight.
`edge-customizations[].options.angle`	`angle`	`25deg`	How sharply a bent edge curves. Ignored when `bend` is `false` .
`edge-customizations[].options.label`	`content` / `dictionary`	none	Tree edge label content, or graph edge-label text overrides. A dictionary can include `content` for trees.

Nested keys for `edge-customizations[].options.label`

Path	Type	Default	Description
`edge-customizations[].options.label.size`	`length`	`style.label-text.size`	Edge-label size.

`edge-customizations[].options.label.content`	`content`	tree only	Tree edge label content. Graph edge labels come from adjacency entries such as `("B", [4])` .
`edge-customizations[].options.label.color`	`color`	`style.label-text.color`	Edge-label text color.
`edge-customizations[].options.label.rotation`	`angle` / `str`	`style.label-text.rotation`	An angle, or `"edge"` to follow the tree/graph edge angle.
`edge-customizations[].options.label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`edge-customizations[].options.label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `node-customizations`

Path	Type	Default	Description
`node-customizations[]`	`tuple`	n/a	One `(node, options)` tuple.
`node-customizations[].node`	`content`	required	Hand-composed node label.
`node-customizations[].options`	`dictionary`	required	Per-node drawing options.
`node-customizations[].options.fill`	`color`	normal fill	Node fill.
`node-customizations[].options.stroke`	`stroke`	normal stroke	Node outline.
`node-customizations[].options.shape`	`str`	`style.node-shape`	Trees/graphs only: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`node-customizations[].options.node-radius`	`float`	`style.node-radius`	Trees/graphs only. Shape size.
`node-customizations[].options.text`	`dictionary`	`style.node-text`	Text style merged into the node text.

Nested keys for `node-labels`

Path	Type	Default	Description
`node-labels[]`	`tuple`	n/a	One `(node, label)` tuple.
`node-labels[].node`	`content`	required	Hand-composed node label.
`node-labels[].label`	`content` / `dictionary`	required	Bare content for quick labels, or a dictionary with `content` plus style and placement keys.
`node-labels[].label.content`	`content`	required for dict labels	Label body. `body` is also accepted.
`node-labels[].label.position`	`str` / `angle`	`"right"`	`"right"` , `"left"` , `"top"` , `"bottom"` , or an angle.
`node-labels[].label.offset`	`point`	`(0, 0)`	Extra `(dx, dy)` shift after the position is applied.
`node-labels[].label.size`	`length`	`style.label-text.size`	Node-label size.
`node-labels[].label.color`	`color`	`style.label-text.color`	Node-label text color.

`node-labels[].label.rotation`	`angle`	`style.label-text.rotation`	Rotation of the label text itself.
`node-labels[].label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`node-labels[].label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `style.node-labels`

Path	Type	Default	Description
`style.node-labels.position`	`str` / `angle`	`"right"`	Default position for all node labels in this diagram.
`style.node-labels.offset`	`point`	`(0, 0)`	Default extra `(dx, dy)` shift for all node labels.
`style.node-labels.gap`	`float`	`0.22`	Default distance from the node boundary in canvas units.
`style.node-labels.size`	`length`	`style.label-text.size`	Default node-label text size.
`style.node-labels.color`	`color`	`style.label-text.color`	Default node-label text color.
`style.node-labels.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-labels.weight`	`str` / `int`	inherits	Typst text weight.
`style.node-labels.rotation`	`angle`	`style.label-text.rotation`	Default rotation of the label text itself.

`node(label, left:, right:, children:, fill:)` : a node with any content label. Use `left` / `right` for binary trees, or `children` for a multiway tree such as a B-tree schematic.

Argument	Type	Default	Description
`label`	`content`	(required)	The node's label; any content, not just a key.
`left`	`none` / `node()` / `subtree()`	`none`	Left child.
`right`	`none` / `node()` / `subtree()`	`none`	Right child.
`children`	`none` / `array`	`none`	Multiway children. When set, this replaces `left` / `right` for layout and rendering.
`fill`	`none` / `color`	`none`	Fill tint for this node.

Nested keys for `node()` children

Path	Type	Default	Description
`left.label`	`content`	required when `left` is `node()` or `subtree()`	Label drawn inside the left child.
`left.left`	`none` / `node()` / `subtree()`	`none`	Nested left child.
`left.right`	`none` / `node()` / `subtree()`	`none`	Nested right child.
`right.label`	`content`	required when `right` is `node()` or `subtree()`	Label drawn inside the right child.
`right.left`	`none` / `node()` / `subtree()`	`none`	Nested left child.
`right.right`	`none` / `node()` / `subtree()`	`none`	Nested right child.
`children[]`	`node()` / `subtree()`	n/a	One child in a multiway tree, laid out left to right.

`subtree(label, fill:, height:, scale:)` : a triangle leaf standing in for an elided subtree.

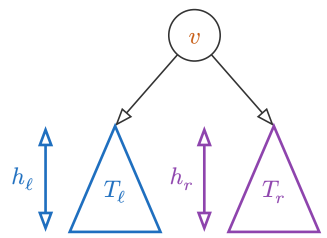
Argument	Type	Default	Description
`label`	`content`	(required)	Label drawn inside the triangle.
`fill`	`none` / `color`	`none`	Outline and label tint.
`height`	`none` / `content`	`none`	Optional side bracket labeling the subtree's height.
`scale`	`float`	`1`	Resizes the triangle relative to the theme's `tri-w` / `tri-h` .

Style keys used by `subtree()`

Path	Type	Default	Description
`style.tri-w`	`float`	`1.2`	Triangle base width before `subtree(scale:)` .
`style.tri-h`	`float`	`1.4`	Triangle height before `subtree(scale:)` .
`style.node-text`	`dictionary`	`(size: 9pt)`	Triangle label and height-bracket text styling.

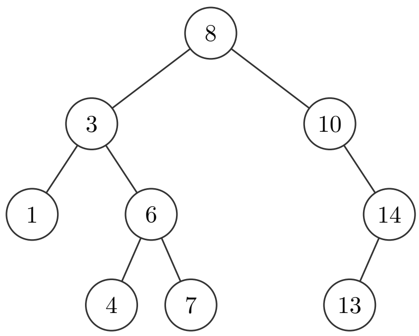
Typst

```
#tree(
  node(
    text(fill: rgb("#C75B12"))[$v$],
    left: subtree($T_ell$, fill: rgb("#1F6FBF"), height: $h_ell$),
    right: subtree($T_r$, fill: rgb("#8E44AD"), height: $h_r$),
  ),
  style: (edge-arrow: true),
)
```



Nest `node(...)` calls to draw a full manual tree. The indentation below mirrors the tree shape: each child is written inside its parent's `left:` or `right:` argument.

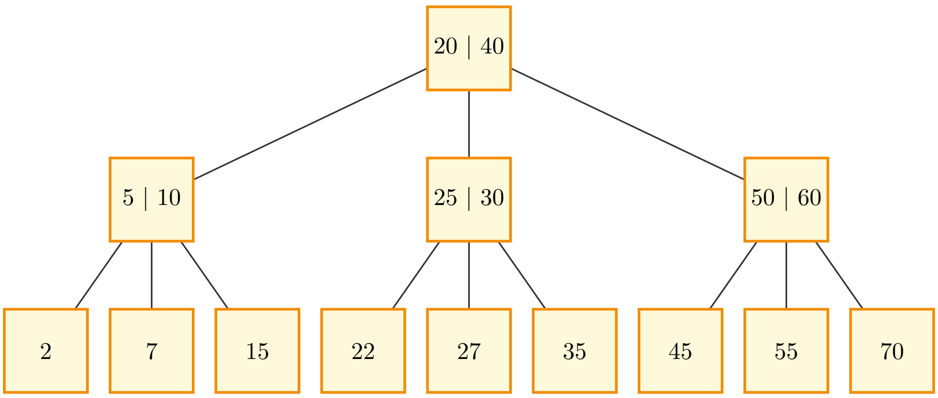
```typst
#tree(
  node(
    [8],
    left: node(
      [3],
      left: node([1]),
      right: node(
        [6],
        left: node([4]),
        right: node([7]),
      ),
    ),
    right: node(
      [10],
      right: node([14], left: node([13])),
    ),
  ),
)
```



Use `children: (...)` for B-tree-style or n-ary schematics.

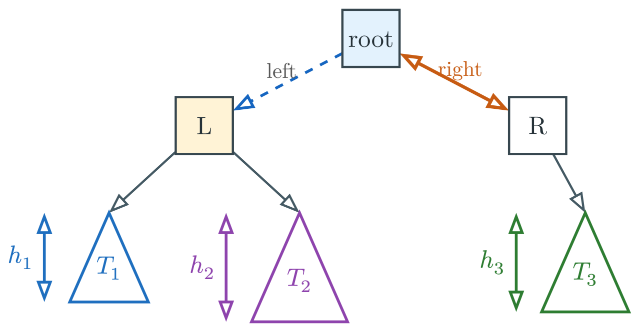
Typst

```
#tree(
  node([20 | 40], children: (
    node([5 | 10], children: (
      node([2]),
      node([7]),
      node([15]),
    )),
    node([25 | 30], children: (
      node([22]),
      node([27]),
      node([35]),
    )),
    node([50 | 60], children: (
      node([45]),
      node([55]),
      node([70]),
    )),
  )),
  style: (
    node-shape: "square",
    node-radius: 0.55,
    node-fill: rgb("#FFF9DB"),
    node-stroke: 1pt + rgb("#F08C00"),
    x-gap: 1.4,
    y-gap: 2.0,
  ),
)
```



Argument coverage: hand-composed trees combine nested `node()` calls, `subtree()` leaves, per-node fills, subtree height labels, tree style, and edge customizations keyed by the visible node labels.

```typst
#tree(
  node(
    [root],
    fill: rgb("#E3F2FD"),
    left: node(
      [L],
      fill: rgb("#FFF3D6"),
      left: subtree($T_1$, fill: rgb("#1F6FBF"), height: $h_1$, scale: 0.9),
      right: subtree($T_2$, fill: rgb("#8E44AD"), height: $h_2$, scale: 1.1),
    ),
    right: node(
      [R],
      right: subtree($T_3$, fill: rgb("#2E7D32"), height: $h_3$),
    ),
  ),
  style: (
    node-shape: "square",
    node-radius: 0.36,
    x-gap: 1.2,
    y-gap: 1.1,
    tri-w: 1.1,
    tri-h: 1.25,
    node-stroke: 0.8pt + rgb("#37474F"),
    node-text: (size: 9pt, color: rgb("#263238")),
    edge-stroke: 0.8pt + rgb("#455A64"),
    edge-arrow: "end",
    scale: 1.05,
  ),
  edge-customizations: (
    ([root], [L], (pattern: "dashed", color: rgb("#1565C0"), label: [left])),
    ([root], [R], (stroke: 1.2pt + rgb("#C75B12"), arrow: "both", label: (content:
    [right], color: rgb("#C75B12")))),
  ),
)
```



5 Operation transitions

`transition` is one-shot: build the structure from `keys`, apply the operation, and render the before state, an arrow, and the derived after state with the diff highlighted, in one call, no variable required. To keep operating on a structure across several calls, use the object model in Section 6 instead.

```typst
#transition(variant, keys, op, style: (:), edge-customizations: (), node-customizations: (), node-labels: (:))
```

Argument	Type	Default	Description
`variant`	`str`	(required)	`"bst"`, `"avl"`, `"min-heap"`, or `"max-heap"`.
`keys`	`array`	(required)	Keys the structure is built from, inserted in order.
`op`	operation	(required)	`tree-insert(key)`, `tree-delete(key)`, or `tree-search(key)` for trees. `heap-insert(key)` or `heap-extract` for heaps.
`style`	`dictionary`	`(:)`	Per-call style override merged over the defaults. See Section 7.
`edge-customizations`	`array`	`()`	Tree transition only. `(parent, child, options)` tuples restyling individual tree edges by value.
`node-customizations`	`array`	`()`	Tree transition only. `(node, options)` tuples restyling individual tree nodes by value.
`node-labels`	`array` / `dictionary`	`(:)`	Tree transition only. Labels drawn outside individual tree nodes.

Nested keys for `style`

Path	Type	Default	Description
`style.node-radius`	`float`	`0.34`	Circle radius, or half the square/diamond/hexagon size, in canvas units.
`style.node-shape`	`str`	`"circle"`	`"circle"`, `"square"`, `"diamond"`, or `"hexagon"`.
`style.x-gap`	`float`	`1.05`	Horizontal spacing between in-order columns.
`style.y-gap`	`float`	`1.2`	Vertical spacing between depths.
`style.node-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Node outline.
`style.node-fill`	`color`	`white`	Default node fill.
`style.edge-stroke`	`stroke`	`0.6pt + rgb("#333333")`	Parent-child edge stroke.
`style.edge-arrow`	`none` / `bool` / `str`	`none`	Edge arrowheads: `none` / `false`, `"end"` or `true`, `"start"`, or `"both"`.
`style.edge-pattern`	`str`	`"normal"`	`"normal"`, `"dashed"`, `"dotted"`, or `"wavy"`.

`style.edge-wave-amplitude`	`float`	`0.07`	Wave height in canvas units.
`style.edge-wave-step`	`float`	`0.14`	Approximate wavelength in canvas units.
`style.node-text`	`dictionary`	`(size: 9pt)`	Node text dictionary. See nested keys below.
`style.label-text`	`dictionary`	`(fill: rgb("#555555"))`	Annotation text dictionary. See nested keys below.
`style.scale`	`float`	`1.0`	Uniform scale on `.diagram` , `.before` , and `.after` .
`style.diff-colors`	`bool`	`true`	`false` keeps operation marks but uses normal fills.

Nested keys for `style.node-text`

Path	Type	Default	Description
`style.node-text.size`	`length`	inherits	Text size.
`style.node-text.color`	`color`	inherits	Text color.
`style.node-text.rotation`	`angle`	`0deg`	Text rotation applied by the diagram renderer.
`style.node-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for `style.label-text`

Path	Type	Default	Description
`style.label-text.size`	`length`	85% of `style.node-text.size`	Label text size.
`style.label-text.color`	`color`	`rgb("#555555")`	Label text color.
`style.label-text.rotation`	`angle` / `str`	`0deg`	Label rotation. Tree and graph edge labels also accept `"edge"` to follow the edge angle.
`style.label-text.font`	`str` / `array`	inherits	Typst text font selection.
`style.label-text.weight`	`str` / `int`	inherits	Typst text weight.

Nested keys for diff-highlight dictionaries

Path	Type	Default	Description
`style.new-style`	`color` / `dictionary`	`rgb("#C8E6C9")`	Node/cell added by an operation. A color is shorthand for `(fill: color)` .
`style.path-style`	`color` / `dictionary`	`rgb("#FFE9A8")`	Traversal or sift path.
`style.remove-style`	`color` / `dictionary`	`rgb("#FFCDD2")`	Node/cell removed by an operation.
`style.rotate-style`	`color` / `dictionary`	`rgb("#BBDEFB")`	AVL rotation nodes and visible subtree roots.

`style.*-style.fill`	`color`	mark default	Fill for the marked node/cell.
`style.*-style.shape`	`str`	`style.node-shape`	Marked tree/heap node shape: `"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`style.*-style.stroke`	`stroke`	normal stroke	Marked node/cell outline.
`style.*-style.node-radius`	`float`	`style.node-radius`	Marked tree/heap node radius.
`style.*-style.text`	`dictionary`	`style.node-text`	Marked node/cell text overrides. Accepts `size` , `color` , `font` , and `weight` .

Nested keys for `edge-customizations`

Path	Type	Default	Description
`edge-customizations[]`	`tuple`	n/a	One `(from, to, options)` tuple. For trees, `from` / `to` are parent and child values; for graphs, node labels.
`edge-customizations[].from`	`content`	required	Source node identity.
`edge-customizations[].to`	`content`	required	Target node identity.
`edge-customizations[].options`	`dictionary`	required	Per-edge options.
`edge-customizations[].options.stroke`	`stroke`	none	Full stroke override. Wins over `color` if both are set.
`edge-customizations[].options.color`	`color`	none	Paint only, at the default thickness.
`edge-customizations[].options.pattern`	`str`	`"normal"`	`"normal"` , `"dashed"` , `"dotted"` , or `"wavy"` .
`edge-customizations[].options.arrow`	`none` / `bool` / `str`	none	Per-edge arrowheads: `none` , `"end"` / `true` , `"start"` , or `"both"` .
`edge-customizations[].options.bend`	`str` / `bool`	`false`	`"left"` or `"right"` curves the edge relative to its travel direction. `false` keeps it straight.
`edge-customizations[].options.angle`	`angle`	`25deg`	How sharply a bent edge curves. Ignored when `bend` is `false` .
`edge-customizations[].options.label`	`content` / `dictionary`	none	Tree edge label content, or graph edge-label text overrides. A dictionary can include `content` for trees.

Nested keys for `edge-customizations[].options.label`

Path	Type	Default	Description
`edge-customizations[].options.label.size`	`length`	`style.label-text.size`	Edge-label size.
`edge-customizations[].options.label.content`	`content`	tree only	Tree edge label content. Graph edge labels come from adjacency entries such as `("B", [4])` .

`edge-customizations[].options.label.color`	`color`	`style.label-text.color`	Edge-label text color.
`edge-customizations[].options.label.rotation`	`angle` / `str`	`style.label-text.rotation`	An angle, or `"edge"` to follow the tree/graph edge angle.
`edge-customizations[].options.label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.
`edge-customizations[].options.label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `node-customizations`

Path	Type	Default	Description
`node-customizations[]`	`tuple`	n/a	One `(node, options)` tuple.
`node-customizations[].node`	`content`	required	Tree node value.
`node-customizations[].options`	`dictionary`	required	Per-node drawing options.
`node-customizations[].options.fill`	`color`	normal fill	Node fill.
`node-customizations[].options.stroke`	`stroke`	normal stroke	Node outline.
`node-customizations[].options.shape`	`str`	`style.node-shape`	Trees/graphs only: `"circle"`, `"square"`, `"diamond"`, or `"hexagon"`.
`node-customizations[].options.node-radius`	`float`	`style.node-radius`	Trees/graphs only. Shape size.
`node-customizations[].options.text`	`dictionary`	`style.node-text`	Text style merged into the node text.

Nested keys for `node-labels`

Path	Type	Default	Description
`node-labels[]`	`tuple`	n/a	One `(node, label)` tuple.
`node-labels[].node`	`content`	required	Tree node value.
`node-labels[].label`	`content` / `dictionary`	required	Bare content for quick labels, or a dictionary with `content` plus style and placement keys.
`node-labels[].label.content`	`content`	required for dict labels	Label body. `body` is also accepted.
`node-labels[].label.position`	`str` / `angle`	`"right"`	`"right"`, `"left"`, `"top"`, `"bottom"`, or an angle.
`node-labels[].label.offset`	`point`	`(0, 0)`	Extra `(dx, dy)` shift after the position is applied.
`node-labels[].label.size`	`length`	`style.label-text.size`	Node-label size.
`node-labels[].label.color`	`color`	`style.label-text.color`	Node-label text color.
`node-labels[].label.rotation`	`angle`	`style.label-text.rotation`	Rotation of the label text itself.
`node-labels[].label.font`	`str` / `array`	`style.label-text.font`	Typst text font selection.

`node-labels[].label.weight`	`str` / `int`	`style.label-text.weight`	Typst text weight.

Nested keys for `style.node-labels`

Path	Type	Default	Description
`style.node-labels.position`	`str` / `angle`	`"right"`	Default position for all node labels in this diagram.
`style.node-labels.offset`	`point`	`(0, 0)`	Default extra `(dx, dy)` shift for all node labels.
`style.node-labels.gap`	`float`	`0.22`	Default distance from the node boundary in canvas units.
`style.node-labels.size`	`length`	`style.label-text.size`	Default node-label text size.
`style.node-labels.color`	`color`	`style.label-text.color`	Default node-label text color.
`style.node-labels.font`	`str` / `array`	inherits	Typst text font selection.
`style.node-labels.weight`	`str` / `int`	inherits	Typst text weight.
`style.node-labels.rotation`	`angle`	`style.label-text.rotation`	Default rotation of the label text itself.

The highlight colors are: yellow for the traversal path, green for an added node, red for a removed node, and blue for the nodes and visible subtree roots in an AVL rotation schematic.

5.1 Tree operations: `tree-insert()`, `tree-delete()`, `tree-search()`

Each is a constructor. Call it with a key and it returns the closure `transition` applies. These aren't object methods. Section 6 covers the `.insert` / `.delete` / `.search` fields on a `bst` / `avl` object, which wrap these same operations.

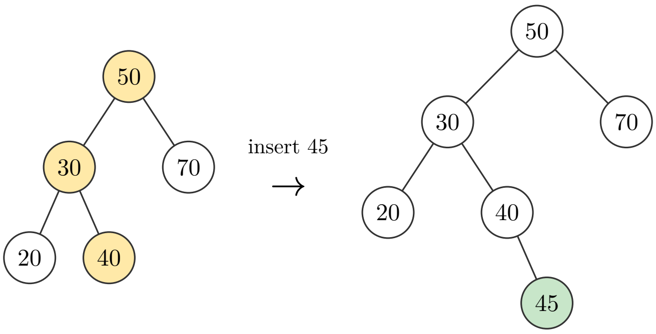
Argument	Type	Default	Description
`key`	`int` / `content`	(required)	The key to insert, delete, or search for.
`rebalance`	`dictionary`	`(:)`	`tree-insert` only. `(enabled:, all-steps:)` , both `false` by default. See below.
`step-label`	`content`	auto	Overrides the arrow label for this operation step.

Nested keys for `rebalance`

Path	Type	Default	Description
`rebalance.enabled`	`bool`	`false`	Show the unrotated AVL tree before a rotation.
`rebalance.all-steps`	`bool`	`false`	For double rotations, split the inner and outer rotations into separate panels.

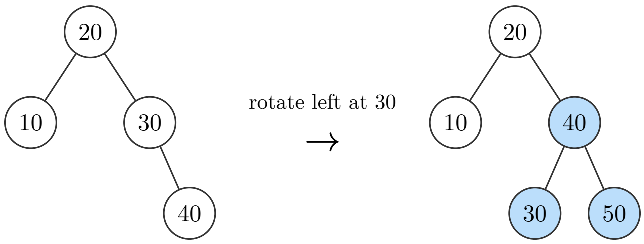
A BST insert highlights the search path on the before state and the new node on the after state.

```
#transition("bst", (50, 30, 70, 20, 40), tree-insert(45))
```



An AVL insert that unbalances the tree triggers a rotation. Every node that changed parent turns blue: the pivot **and** the child promoted above it, plus any subtree that swapped from one to the other. The new node is green.

```
#transition("avl", (20, 10, 30, 40), tree-insert(50))
```

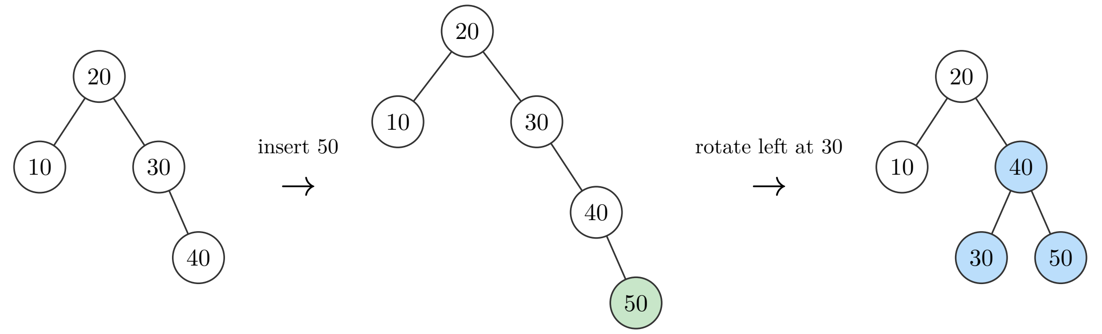


Argument	Type	Default	Description
`enabled`	`bool`	`false`	Adds a panel for the tree right after the plain insert, before any rotation straightens it out, when the AVL fixup actually rotates. A BST insert, or an AVL insert that never unbalances the tree, ignores this and stays 2-panel.
`all-steps`	`bool`	`false`	A **double** rotation (left-right or right-left) normally collapses straight from the unbalanced tree to the final one. `true` splits it into its own inner-then-outer panels instead. No effect on a single rotation, since there's only one step either way.

`rebalance: (enabled: true)` splits that same rotation into three panels: the tree right after the plain insert (unbalanced, new node green), then the rotation that fixes it (every reparented node blue).

```typst
#transition("avl", (20, 10, 30, 40), tree-insert(50, rebalance: (enabled:
true)))
```

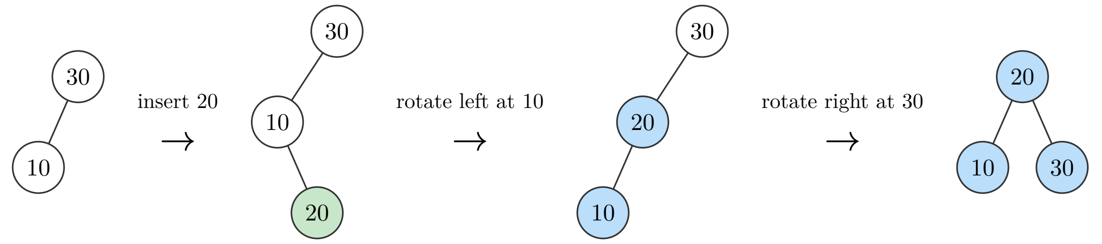
Typst



A double rotation, left-right or right-left, needs two rotations to fix. By default `enabled: true` still collapses it to 3 panels, marking every node either rotation touched; `all-steps: true` shows the inner rotation as its own step, for 4 panels total.

```typst
#transition("avl", (30, 10), tree-insert(20, rebalance: (enabled: true, all-
steps: true)))
```

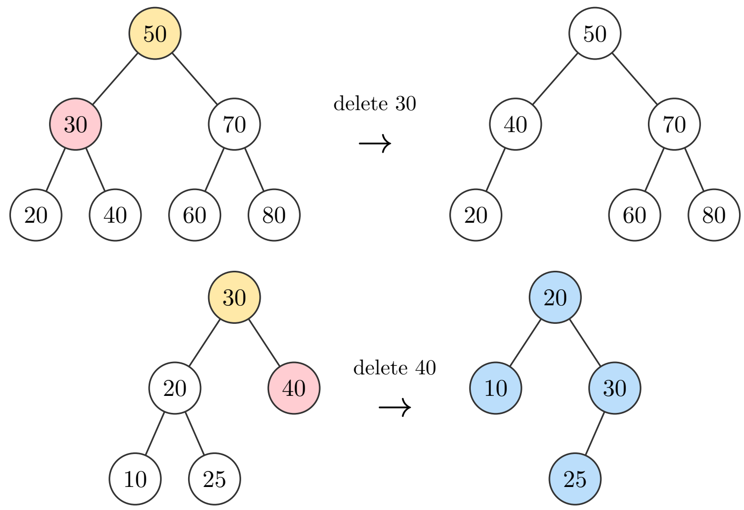
Typst



A delete marks the search path on the before state and gives the removed node the remove style. Two-child nodes are replaced by their in-order successor; AVL deletes also rebalance when removal makes a subtree too heavy, marking rotation nodes blue.

```
#transition("bst", (50, 30, 70, 20, 40, 60, 80), tree-delete(30))
#transition("avl", (30, 20, 40, 10, 25), tree-delete(40))
```

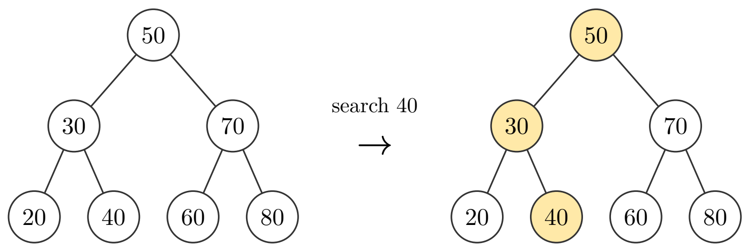
Typst



A search leaves the before state plain and highlights the traversal path on the after state.

```
#transition("bst", (50, 30, 70, 20, 40, 60, 80), tree-search(40))
```

Typst



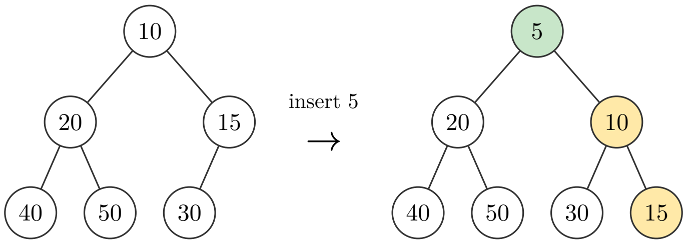
5.2 Heap operations: `heap-insert()` and `heap-extract`

`heap-insert` is a constructor, like `tree-insert` . `heap-extract` is different: it takes no key (a heap can only pull its root), so it **is** the operation directly, passed without calling it. Object methods such as `(heap.extract)(step-label: [...])` can still override a step label.

Argument	Type	Default	Description
`key`	`int` / `content`	(required)	The key to insert, `heap-insert` only. `heap-extract` takes nothing.
`step-label`	`content`	auto	Overrides the arrow label. For `heap-extract` , use it on the object method: `(h.extract)(step-label: [...])` .

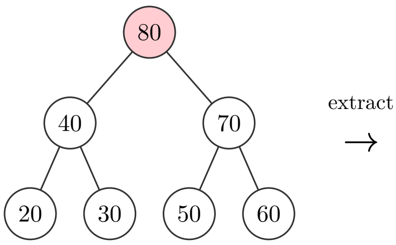
`heap-insert` appends the key, marks the inserted value green, and highlights the existing sift-up path in yellow:

```typst
#transition("min-heap", (10, 20, 15, 40, 50, 30), heap-insert(5))
```



`heap-extract` removes the root (marked red), moves the last leaf into its place, and highlights the sift-down path in yellow:

```typst
#transition("max-heap", (50, 30, 70, 20, 40, 60, 80), heap-extract)
```



> **Note.** Heap operation marks are keyed by array position while rendering, so inserting a value already present in the heap can still highlight only the new node. Tree operation marks remain value-keyed; see Section 9.

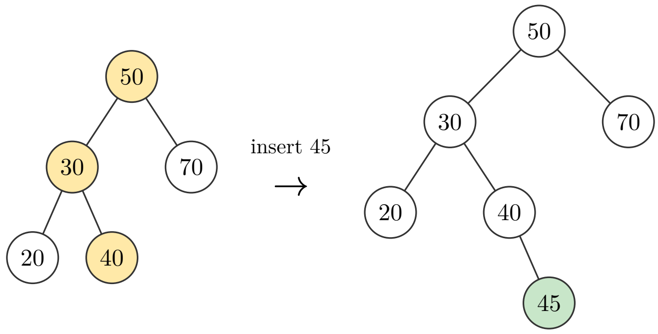
6 Objects and operations

6.1 The object model

Every builder in Section 3 returns a dictionary that is both the drawing and the structure you operate on. `.diagram` holds the rendering. An operation field derives the next state: `.insert` / `.delete` / `.search` on trees, `.insert` / `.extract` on heaps, `.push` / `.pop` on stacks, `.enqueue` / `.dequeue` on queues, `.insert` / `.delete` on linked lists and doubly linked lists.

Typst requires parentheses around a callable dictionary field, so calling an operation looks like `(obj.insert)(45)` , not `obj.insert(45)` .

```typst
#let b = bst(50, 30, 70, 20, 40)
#(b.insert)(45).diagram
```



Calling an operation field returns a **step**: a dictionary with five fields.

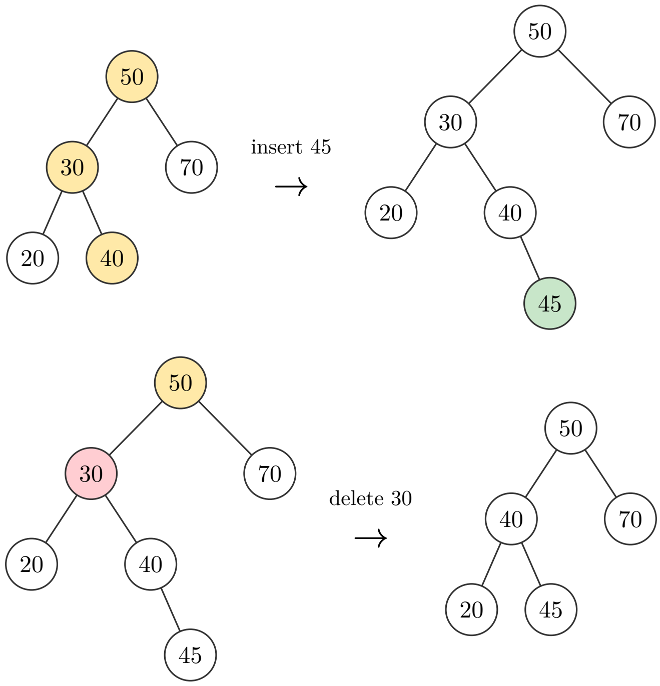
Field	Description
`.label`	The arrow's text, e.g. `"insert 45"` or `"extract"` .
`.before`	Rendered picture of the state **before** the operation, with that side's highlight marks.
`.after`	Rendered picture of the state **after** the operation, with that side's highlight marks. A picture only: no `.insert` / `.push` /etc. of its own.
`.diagram`	The packaged `.before` → arrow → `.after` row, what you show by default.
`.result`	The **next live object**, same shape as what the builder returned: `.diagram` plus operation fields, ready for the next call.

Note. `.result` is fully displayable via `result.diagram` , but plain, with no highlights. Once you have `.result` , it is a fresh object built from the post-operation state, exactly as if you had called `bst(...)` (or `min-heap(...)` , etc.) on those exact keys directly. The diff-marking machinery belongs to the one step that produced it. It doesn't travel with the resulting object, and it doesn't accumulate across a chain of `.result` calls.

6.2 Chaining operations

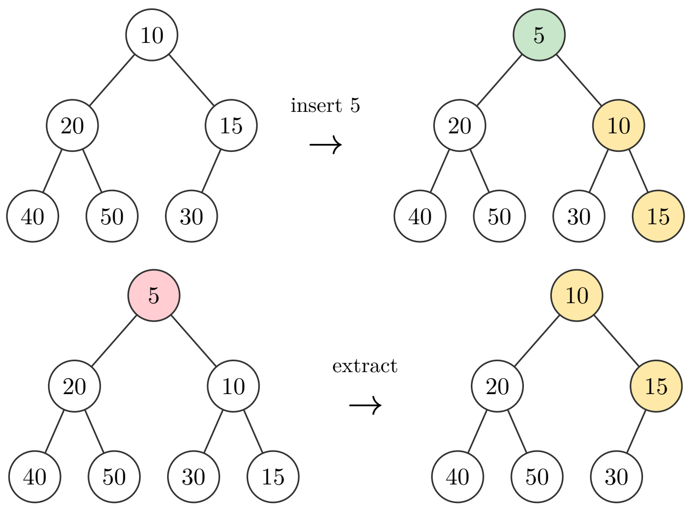
Keep calling operations on `.result` to chain an arbitrary sequence, each with its own fresh highlight:

```typst
#let b = bst(50, 30, 70, 20, 40)
#let step = (b.insert)(45)
#std.stack(spacing: 1em,
  step.diagram,
  ((step.result).delete)(30).diagram,
)
```



Every structure's natural operations chain the same way: heaps (`.insert` / `.extract`), stacks (`.push` / `.pop`), queues (`.enqueue` / `.dequeue`), linked lists (`.insert` / `.delete`):

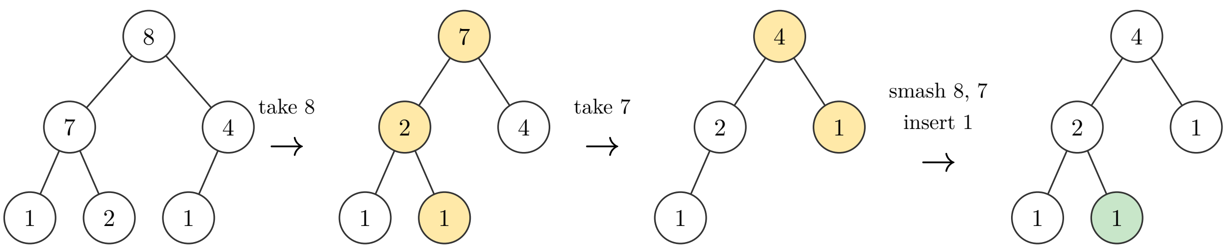
```typst
#let h = min-heap(10, 20, 15, 40, 50, 30)
#let step = (h.insert)(5)
#std.stack(spacing: 1em,
  step.diagram,
  ((step.result).extract)().diagram,
)
```



Pass `step-label:` to an operation method when a sequence needs a problem-specific arrow label:

```typst
#let h = max-heap(2, 7, 4, 1, 8, 1)
#let s1 = (h.extract)(step-label: [take 8])
#let s2 = ((s1.result).extract)(step-label: [take 7])
#let s3 = ((s2.result).insert)(1, step-label: [smash 8, 7 \ insert 1])

#sequence(h, s1, s2, s3, mode: "after", columns: 7)
```



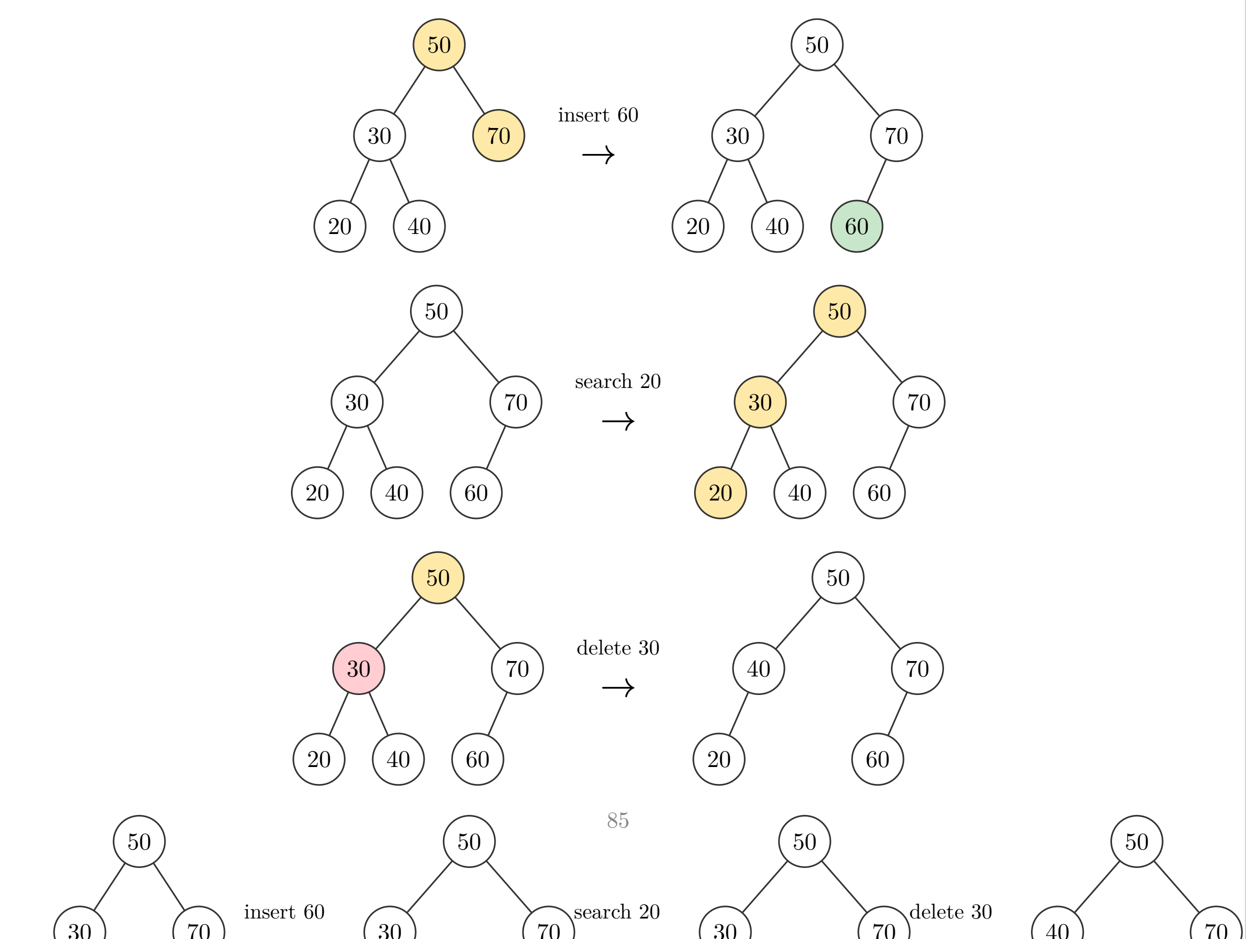
6.3 Wrapped operation sequences

`sequence(..., columns:, mode:)` lays out step objects in a grid. It accepts operation steps directly, so you do not need to write `.diagram` for each one. Use `columns: 1` for a vertical trace, or a larger number to wrap across rows. The default `mode: "all"` shows each full transition row. Compact modes keep operation arrows between states: `mode: "after"` shows highlighted after panels, while `mode: "result"` shows the plain live object after each step, without operation highlights. Put the starting object first when you want the trace to begin with the original structure.

Argument	Type	Default	Description
`..steps`	step / content	(required)	Operation steps, live objects, or already-rendered content.
`columns`	`int`	`1`	Number of grid columns before wrapping. Compact modes count arrows as columns too.
`gap`	`length`	`1em`	Horizontal gap between columns.
`row-gap`	`length`	`1em`	Vertical gap between rows.
`mode`	`str`	`"all"`	`"all"` / `"diagram"` renders `.diagram` compact modes `"before"` , `"after"` , and `"result"` render selected states with labeled arrows between operations.

```typst
#let b = bst(50, 30, 70, 20, 40)
#let s1 = (b.insert)(60)
#let s2 = ((s1.result).search)(20)
#let s3 = ((s2.result).delete)(30)

#sequence(s1, s2, s3, columns: 1, row-gap: 1.2em)
#sequence(b, s1, s2, s3, mode: "result", columns: 7)
```

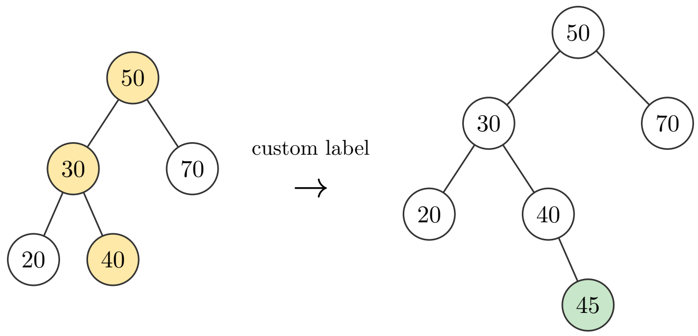


6.4 Custom arrows with `op-arrow()`

To compose the arrow yourself instead of using `.diagram` , combine the step's `.before` / `.after` with `op-arrow(label, symbol:)` .

Argument	Type	Default	Description
`label`	`content`	(required)	Text shown above the arrow.
`symbol`	`content`	`$arrow.r$`	The arrow glyph itself.

Typst
```
#let step = (bst(50, 30, 70, 20, 40).insert)(45)
#std.stack(dir: ltr, spacing: 1.2em,
  align(horizon, step.before),
  op-arrow[custom label],
  align(horizon, step.after),
)
```

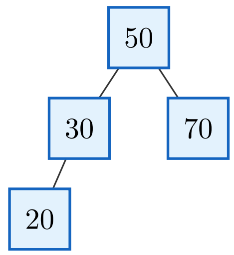


7 Styling

Every builder takes a `style:` dictionary that is merged over the package defaults, so colors, size, and shape can be overridden per call. `theme` is the default dictionary itself and `resolve(style)` is the merge helper (`theme + style`); both are exported if you want to build your own styling utilities on top. Raw dictionaries remain supported.

Typst

```
#bst(50, 30, 70, 20, style: tree-style(
  node-shape: "square",
  node-radius: 0.4,
  node-fill: rgb("#E3F2FD"),
  node-stroke: 1pt + rgb("#1565C0"),
  node-text: text-style(size: 11pt),
)).diagram
```



Note. To set a document-wide default instead of repeating `style:` at every call, rebind the builder once: `#let bst = bst.with(style: tree-style(..))` .

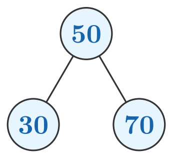
Note. Editor tip: Typst dictionaries do not expose a schema to autocomplete. Use the style-builder functions when you want argument suggestions while writing `style:` . Raw dictionaries still work when you prefer them.

7.1 Autocomplete-friendly style builders

Structure-specific style builders return sparse dictionaries. Because their keys are named function arguments, Typst editors suggest only relevant keys while you type `style: tree-style(...)` , `style: stack-style(...)` , and so on. Omitted arguments do not overwrite package or structure-specific defaults. Nested builders provide the same completion for dictionary-valued settings.

Typst

```
#bst(
  50, 30, 70,
  style: tree-style(
    x-gap: 1.4,
    node-fill: rgb("#E7F5FF"),
    node-text: text-style(
      size: 10pt,
      color: rgb("#1864AB"),
      weight: "bold",
    ),
    new-style: node-mark-style(
      fill: rgb("#D3F9D8"),
      stroke: 1pt + rgb("#2B8A3E"),
    ),
  ),
).diagram
```



Argument	Type	Default	Description
`tree-style(...)`	`function`	n/a	Styles BSTs, AVL trees, manual trees, and tree transitions.
`heap-style(...)`	`function`	n/a	Styles min/max heaps and heap transitions.
`graph-style(...)`	`function`	n/a	Styles graphs.
`list-style(...)`	`function`	n/a	Styles linked and doubly linked lists.
`stack-style(...)`	`function`	n/a	Styles stacks.
`queue-style(...)`	`function`	n/a	Styles queues.
`array-style(...)`	`function`	n/a	Styles arrays, including `indices:` .
`matrix-style(...)`	`function`	n/a	Styles matrices.
`text-style(size:, color:, fill:, font:, weight:, rotation:)`	`function`	n/a	Builds `style.node-text` or mark text overrides.
`label-style(size:, color:, fill:, font:, weight:, rotation:)`	`function`	n/a	Builds `style.label-text` .

`node-mark-style(fill:, shape:, stroke:, node-radius:, text:)`	`function`	n/a	Builds tree/heap diff styles.
`cell-mark-style(fill:, stroke:, text:)`	`function`	n/a	Builds linear-structure diff styles without unsupported node geometry.
`node-label-style(position:, offset:, gap:, size:, color:, font:, weight:, rotation:)`	`function`	n/a	Builds `style.node-labels` .
`indices-style(enabled:, labels:, offset:, size:, color:, font:, weight:)`	`function`	n/a	Builds `style.indices` . Pass `labels: auto` for zero-based indices.

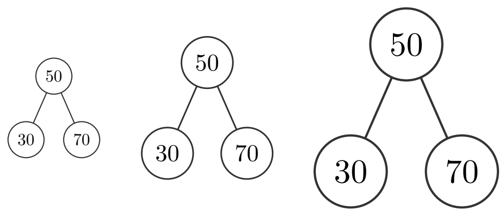
7.2 Where keys are documented

The detailed key lists now live where the argument appears, including nested dictionaries such as `style.node-text` , `style.label-text` , and `edge-customizations[].options.label` .

Area	Detailed reference
Trees	`bst` , `avl` , `tree` , and `transition` list their full `style.*` , `edge-customizations.*` , `node-customizations.*` , and `node-labels.*` keys in their argument reference.
Heaps	`min-heap` , `max-heap` , and heap transitions list the tree-renderer `style.*` keys they use.
Graphs	`graph` lists `adjacency.*` , `labels.*` , `positions.*` , `edge-customizations.*` , `node-customizations.*` , `node-labels.*` , `edge-customizations[].options.label.*` , and `style.*` .
Linear structures	`linked-list` , `doubly-linked-list` , `stack` , and `queue` list their cell, pointer, label, scale, and diff-highlight `style.*` keys.

Typst

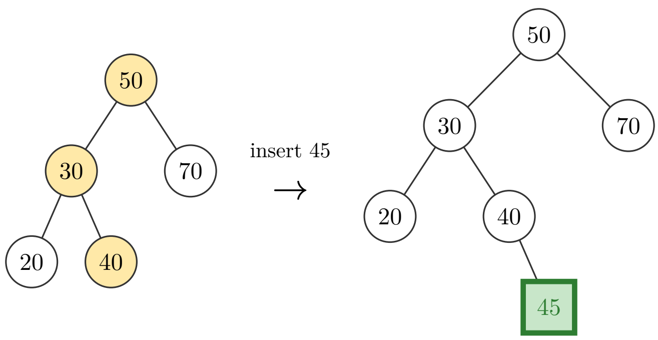
```
#std.stack(dir: ltr, spacing: 1.5em,
  bst(50, 30, 70, style: (scale: 0.7)).diagram,
  bst(50, 30, 70).diagram,
  bst(50, 30, 70, style: (scale: 1.4)).diagram,
)
```



7.3 Diff-highlight styling

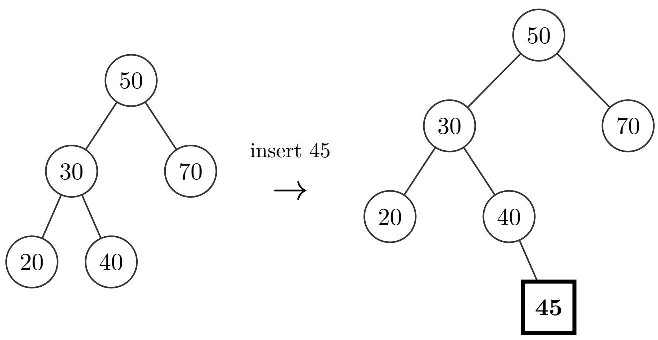
A plain color is shorthand for `(fill: color)` every example so far used that shorthand. Pass a dictionary instead to also override `shape` , `stroke` , `node-radius` , or `text` on just the marked node/cell. Fields the dictionary leaves out fall back to the surrounding `node-shape` / `node-stroke` / `node-radius` , **except** `fill` , which falls back to the key's own default highlight color, not `node-fill` . `text` merges over `node-text` , so each mark kind can have its own label color, size, font, or weight.

```typst
#transition("bst", (50, 30, 70, 20, 40), tree-insert(45), style: (
  // square instead of circle, thicker stroke; fill stays the default green
  new-style: (shape: "square", stroke: 2pt + rgb("#2E7D32"), text: (color:
  rgb("#2E7D32"))),
))
```



Set `diff-colors: false` to remove highlight fill colors while keeping operation marks. Shape, stroke, and radius overrides still apply.

```typst
#transition("bst", (50, 30, 70, 20, 40), tree-insert(45), style: (
  diff-colors: false,
  new-style: (shape: "square", stroke: 1.5pt + black, text: (weight: "bold")),
))
```



> **Note.** This applies to every structure that highlights marks: `bst` / `avl` (node `shape` / `stroke` / `node-radius`), `min-heap` / `max-heap` (same, since they reuse the tree renderer), and `stack` / `queue` / `linked-list` / `doubly-linked-list` (cell `fill` / `stroke` cells have no `shape` or `node-radius` to override).

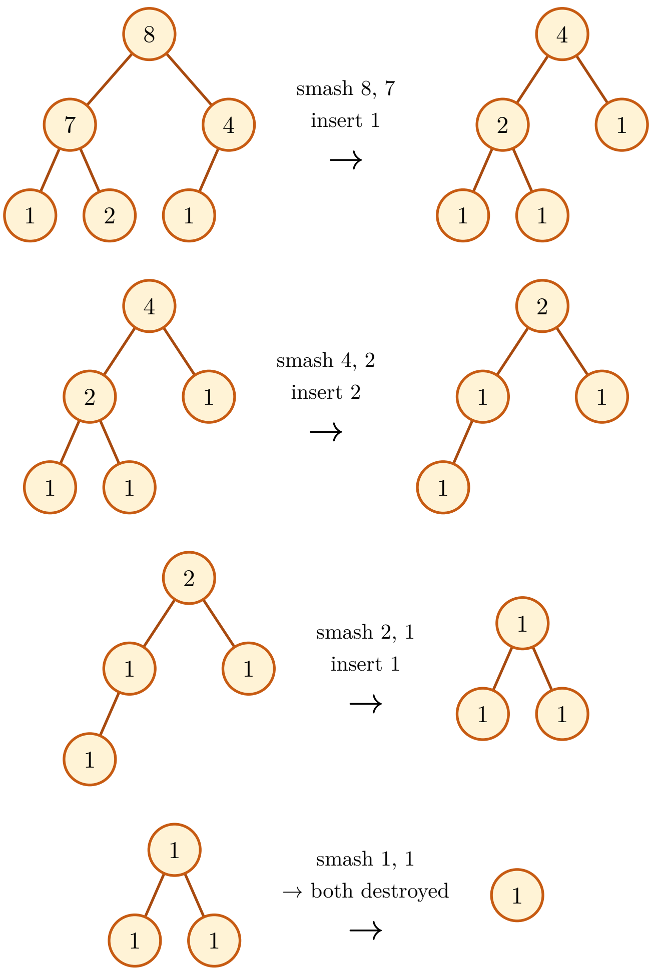
8 Worked example: Last Stone Weight

A complete example pairing a real solution with the diagram it produces: LeetCode 1046, Last Stone Weight (Easy). Repeatedly smash the two heaviest stones together; if they're equal both are destroyed, otherwise the difference goes back in. A max-heap hands you the two heaviest in $O(\log n)$ each round instead of a full rescan.

```python
import heapq

def lastStoneWeight(stones):
    heap = [-s for s in stones]
    heapq.heapify(heap)
    while len(heap) > 1:
        y, x = -heapq.heappop(heap), -heapq.heappop(heap)
        if y != x:
            heapq.heappush(heap, -(y - x))
    return -heap[0] if heap else 0
```

The diagram below runs the **same** algorithm through `max-heap`, chained with `.extract()` / `.insert()` and a custom `op-arrow` label per round. The diagram comes from running the algorithm, not from hand-drawing pictures to match the code:



One stone remains, weight **1**, matching `lastStoneWeight([2, 7, 4, 1, 8, 1]) == 1`.

9 Limitations

- Tree operation diff marks are keyed by node **value**, not position, so a tree with duplicate keys can highlight more nodes than actually moved. Heap operation marks are position-keyed. For static tree/graph diagrams, use `node-customizations` for explicit per-node control. For arrays and matrices, use `cell-customizations` .
- `graph` lays nodes out on a circle. A layout that pulls connected nodes closer together isn't built yet.
- `graph` has no operations yet: no add-node, add-edge, or a transition to match trees and heaps.
- `sequence` wraps already-created operation steps, but it does not yet compute an operation chain from a declarative list by itself.
- Hash tables sit outside this release's scope.

10 Quick reference

10.1 Structure builders

Call	Result
`bst(..keys, style:, edge-customizations:, node-customizations:, node-labels:)`	Binary search tree built by inserting keys in order; ops `insert` , `delete` , `search`
`avl(..keys, style:, edge-customizations:, node-customizations:, node-labels:)`	AVL tree, rebalanced on every insertion and deletion; same ops as `bst`
`min-heap(..keys, style:)`	Array-backed complete binary tree, smallest key at the root; ops `insert` , `extract`
`max-heap(..keys, style:)`	Array-backed complete binary tree, largest key at the root; ops `insert` , `extract`
`linked-list(..vals, style:, pointer:, addresses:, head:)`	Linked list; `pointer` shows `data | next` cells; ops `insert` , `delete`
`doubly-linked-list(..vals, style:, pointer:, addresses:, head:)`	Doubly linked list; `pointer` shows `prev | data | next` cells; ops `insert` , `delete`
`stack(..vals, style:, top-label:)`	Stack, first argument is the top; ops `push` , `pop`
`queue(..vals, style:, enqueue:, dequeue:, front-label:, rear-label:)`	Queue, first argument is the front; ops `enqueue` , `dequeue`
`array-view(..vals, style:, cell-customizations:, pointers:)`	Static array cells with optional `style.indices` , per-cell overrides, and labelled `pointers` arrows above cells
`matrix(rows, style:, cell-customizations:, row-labels:, column-labels:)`	Static matrix/grid cells with optional row/column labels and per-cell overrides
`graph(adjacency, directed:, labels:, positions:, layout:, radius:, edge-customizations:, node-customizations:, node-labels:, style:)`	Graph from an adjacency dict; circular or manual layout, no ops yet

10.2 Sorting algorithms

Call	Result
`merge-sort(array, order:, labels:)`	One breadth-by-depth divide-and-merge tree from a single input array; returns `.diagram` , `.steps` , and `.result`
`merge-operation(left, right, order:, pointers:, labels:)`	Step-by-step merge of two sorted arrays, with default `i` , `j` , and `i+j` cursor pointers and a progressively filled result array
`partition-step(array, order:, pivot:, pointers:, labels:)`	Partition trace with a middle pivot, or a detailed last-pivot quicksort trace with `i` , `j` , and `pivot` markers

`quick-sort(array, order:, pivot:, labels:)`	Breadth-by-depth partition tree: every active subarray partitions on the same level; `pivot` is `"first"` , `"last"` , or an index
`bubble-sort(array, order:, pointers:, labels:, compare:, swap:)`	Comparison and swap trace; pointers mark positions with index arrows by default
`insertion-sort(array, order:, pointers:, labels:, compare:, swap:)`	Sorted-prefix insertion trace using adjacent swaps and default pointer markers
`selection-sort(array, order:, pointers:, labels:, compare:, current:, minimum:, swap:)`	Selection and swap trace with default pointer markers; red marks `i` , purple marks minimum index `m` , and yellow marks inspected index `j` . When `i = m` , the cell combines red with purple stripes
`sort-sequence(steps, columns:, gap:, row-gap:)`	Custom layout for generated sorting steps

10.3 Hand-composed trees

Call	Result
`tree(root, style:, edge-customizations:, node-customizations:, node-labels:)`	Render a tree built from `node` / `subtree`
`node(label, left:, right:, children:, fill:)`	A binary or multiway node with any content label
`subtree(label, fill:, height:, scale:)`	Triangle leaf with an optional height bracket

10.4 Transitions and operations

Call	Result
`transition(variant, keys, op, style:, edge-customizations:, node-customizations:, node-labels:)`	Before → arrow → derived after for `bst` / `avl` / `min-heap` / `max-heap` , diff highlighted
`tree-insert(key, rebalance:, step-label:)`	Opera-

	tion: insert `key` . `rebalance: (enabled: true)` adds a panel for the unrotated tree, when AVL rotates; `all-steps: true` splits a double rotation into its own inner/outer steps
`tree-delete(key, step-label:)`	Operation: delete `key` AVL rebalances when needed
`tree-search(key, step-label:)`	Operation: highlight

	the path to `key`
`heap-insert(key, step-label:)`	Operation: insert `key` , marks the inserted value and the sift-up path
`heap-extract`	Operation: remove the root, marks it and the sift-down path (no key)
`sequence(..steps, columns:, gap:, row-gap:, mode:)`	Wrap operation steps or diagrams into a grid; `mode` chooses full steps, before/after panels, or

	plain re-sults
`op-arrow(label, symbol:)`	Reusable labeled ar-row for cus-tom oper-ation lay-outs

10.5 Styling keys

Key	Default	Effect
`node-radius`	`0.34`	Circle radius / half shape size.
`node-shape`	`"circle"`	`"circle"` , `"square"` , `"diamond"` , or `"hexagon"` .
`x-gap`	`1.05`	Horizontal spacing between columns.
`y-gap`	`1.2`	Vertical spacing between depths.
`node-stroke`	`0.6pt + #333`	Node outline.
`node-fill`	`white`	Default node fill.
`edge-stroke`	`0.6pt + #333`	Edge stroke.
`edge-arrow`	`none`	Edge arrowheads: `none` , `"end"` / `true` , `"start"` , `"both"` .
`edge-arrow-fill`	`none`	`graph` arrowheads: `none` (open) or a color (solid).
`edge-pattern`	`"normal"`	Global tree/ `graph` edge pattern.
`edge-wave-amplitude`	`0.07`	Wave height in canvas units.
`edge-wave-step`	`0.14`	Approximate wavelength in canvas units.
`tri-w`	`1.2`	Subtree triangle base width.
`tri-h`	`1.4`	Subtree triangle height.
`box-w`	`0.95`	Linear-structure cell width.
`box-h`	`0.7`	Linear-structure cell height.
`box-gap`	`0.55`	Gap between cells/rows.
`box-stroke`	`0.6pt + #333`	Cell outline.
`box-fill`	`white`	Default cell fill.
`ptr-fill`	`#D7ECC9`	Linked-list pointer-cell fill.
`prev-ptr-fill`	`#D7ECC9`	Doubly linked list previous-pointer fill.

`next-ptr-fill`	`#D7ECC9`	Doubly linked list next-pointer fill.
`node-text`	`(size: 9pt)`	Node/cell text styling.
`label-text`	`(fill: #555)`	Annotation/edge text styling.
`scale`	`1.0`	Uniform scale on the whole rendered diagram.
`diff-colors`	`true`	Set `false` to keep mark shapes/strokes but use normal fills.
`new-style`	`#C8E6C9`	Added node/cell; color or `(fill:, shape:, stroke:, node-radius:, text:)` .
`path-style`	`#FFE9A8`	Traversal / sift path; color or dict, as above.
`remove-style`	`#FFCDD2`	Removed node/cell; color or dict, as above.
`rotate-style`	`#BBDEFB`	AVL rotation nodes and visible subtree roots; color or dict, as above.

10.6 Styling helpers

Call	Result
`tree-style(..tree keys)`	Sparse BST/AVL/manual-tree style dictionary
`heap-style(..heap keys)`	Sparse min/max-heap style dictionary
`graph-style(..graph keys)`	Sparse graph style dictionary
`list-style(..list keys)`	Sparse linked-list style dictionary
`stack-style(..stack keys)`	Sparse stack style dictionary
`queue-style(..queue keys)`	Sparse queue style dictionary
`array-style(..array keys)`	Sparse array style dictionary
`matrix-style(..matrix keys)`	Sparse matrix style dictionary
`text-style(size:, color:, fill:, font:, weight:, rotation:)`	Node/cell or mark text dictionary
`label-style(size:, color:, fill:, font:, weight:, rotation:)`	Annotation/edge text dictionary

`node-mark-style(fill:, shape:, stroke:, node-radius:, text:)`	Tree/heap diff-highlight dictionary
`cell-mark-style(fill:, stroke:, text:)`	Linear-cell diff-highlight dictionary
`node-label-style(position:, offset:, gap:, size:, color:, font:, weight:, rotation:)`	Default external node-label dictionary
`indices-style(enabled:, labels:, offset:, size:, color:, font:, weight:)`	Array-index dictionary